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TARGET ASIC Characterization and Optimization for a Fiber-Based Phase Camera

Abstract

The Einstein Telescope (ET) is a planned next-generation gravitational wave observatory designed to detect spacetime distortions with unprecedented sensitivity. Accurate control of the alignment of the interferometer optics requires fast, spatially resolved phase sensing of the laser beam, which can be provided by a fiber-array based phase camera.

A cost efficient solution to digitize these multi-channel analog inputs is the TeV Array Readout Electronics with gigasample per second (GSa/s) sample rate and Event Trigger (TARGET) family of application-specific integrated circuits (ASICs). This thesis characterizes and optimizes the TARGET ASICs for application in the fiber-array based phase camera. Clock jitter from different sources are quantified, calibration, timing and synchronization methods are discussed. In addition, the influence of lower sampling rate on the phase resolution is analyzed. The results demonstrate improvements in data acquisition stability by decreasing the clock jitter and improving the phase resolution through calibration methods and sampling rate reduction and prove that TARGET ASICs are a suitable solution for a fiber-array based phase camera for next-generation gravitational wave detectors.

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1 An Introduction

Newtons universal law of gravitation states that the gravitational force between two objects varies inversely with the square of the distance between them. This was recently confirmed in the large scale by studying galaxy clusters, which obey the inverse square law across hundreds of millions of light-years [1]. Einstein's theory of special relativity showed that speed of light is constant in any reference frame which connected the two observables space and time to one spacetime. In his theory of general relativity Einstein endowed spacetime with curvature and made it dynamical. This provided a covariant theory of gravity, in which every prediction for physical measurements in an inertial frame is invariant under coordinate transformation [2, p.3],[3, sec.2.5]. The effect gravity has on matter is described as gravitational force, which is reinterpreted as the behavior of geodesics in the spacetime [2, p.1-3]. Geodesics are locally length minimizing curves, here in the curvature of spacetime [4]. To be in unison with Einsteins special theory of relativity the gravitational force must be causal. Therefore, any change of a gravitating source must travel no faster than the speed of light, c . This leads to the concept of gravitational radiation. Einstein's calculation of this gravitational radiation lead to a leading-order quadrupole formula for gravitational wave emission. Analogous to the dipole formula of electromagnetic radiation, which are produced by accelerated charges, these gravitational waves arise from asymmetric, accelerated masses [2, p.3], [5]. Through the quadrupolar nature of gravitational waves, they stretch space and time in one direction while compressing in the direction perpendicular to that [6].

Indirect experimental confirmation of the existence of gravitational waves, came from the observation of binary neutron star systems, where the gravitational wave emission is large enough that the system's characteristics change on a short enough timescale to be observed. Russel Hulse's and Joe Taylor's tracked a pulsar for 30 years, showing its orbit is decaying. The results matched the prediction of general relativity for such a decay due to loss of orbital energy and angular momentum by gravitational wave emission [2, p.3].

Starting in the 1970s, Rainer Weiss, Kip Thorne, and Barry Barish formulated the concept to measure gravitational waves with a laser interferometer, which is suitable to detect a differential length change. This led to the LIGO facilities, a Laser Interferometer Gravitational-wave Observatory ((LIGO) [7]) consisting of two facilities in the USA, each made up of two perpendicular 4 km long arms. When a gravitational wave passes through, it stretches one arm while compressing the other. This change results in a differential change in time the light takes in each arm, which is measured. Through the two facilities a direction of the gravitational wave can be estimated [8]. These measurements help understanding some of the most fundamentals laws of physics. Dynamics of large-scale events in the universe like the formation and growth of galaxies and the supermassive black holes at their centers can be explained by the study of gravitational waves [9]. LIGO achieved the first measurement of gravitational waves on 14th September, 2015 [10]. From then LIGO was able to detect gravitational waves from many sources, including black hole binaries and a merger of two neutron stars [8].

These detections opened up a new window with a new perspective to the universe. They

not only proved Einstein's theory of general relativity, but can also help understand the formation of asymmetric binary systems such as primordial black holes and Neutron Stars. These systems can now also be studied with gravitational waves besides electromagnetic waves. The mass, abundance and with enhanced sensitivity also details such as eccentricity, redshift, and spatial inhomogeneities of these black holes can become accessible [11].

In multi-messenger astronomy objects are observed with different methods to detect the different messengers, like light and neutrinos. Gravitational waves add a new perspective. The first multi messenger observation involving gravitational waves happened on August 17, 2017. The observatories LIGO and Virgo, a European gravitational wave observatory, detected the merger of a binary neutron star about 1.3 billion lightyears away. Just 1.7s after the merger was observed, the Fermi and Swift space telescopes observed the resulting gamma-ray burst. This showed, that gravity travels at the same speed as light [12]. As current generation gravitational wave observatories are expected to reach the maximum sensitivity for the current facilities infrastructure by the end of the next decade [13, p.1], new gravitational wave observatories like the Einstein Telescope (ET) are planned [14]. To increase the sensitivity in comparison to existing gravitational wave observatories, more precise optical measurement and noise suppression techniques are required.

There are different kinds of noise sources, limiting the sensitivity of the interferometers. Quantum noise, shot noise and radiation pressure noise come from the quantum nature of photons. Other noise sources include thermal noise and environmental noise [15]. Misalignment of the interferometer mirrors generate Higher-order electromagnetic modes which couple into the fundamental Gaussian mode reducing the interferometers sensitivity [16]. Precise alignment control of the optical cavities is therefore necessary.

A fiber-array-based phase camera offers a way to analyze the laser wavefronts with spatial resolution. This provides a way to monitor and correct misalignments in the laser optics. To measure and digitize each fiber in parallel for digital analysis TARGET ASICs are used. Their capabilities are tested with a prototyping board. This thesis focuses on the synchronization of multiple boards, calibration and sampling behavior and influence on the phase resolution in the laboratory setup for optimization in the fiber-array based phase camera for implementation in the future Einstein Telescope.

This thesis characterizes the TARGET ASICs evaluation boards for scalable readout of a fiber-array based phase camera for misalignment control in third generation gravitational wave detectors. In this thesis the synchronization of multiple boards for wavefront sensing with large fiber arrays was tested. A deeper analysis of the calibration methods, pedestals, dc-transfer function and timebase stability was performed to increase sensitivity. The influence of the sampling rate was analyzed with the introduction of 500 MSa/s sampling rate to quantify its impact on phase resolution. The measurements were performed at the ET laboratory at the Erlangen Centre for Astroparticle Physics (ECAP), using the laser setup with sidebands generated by an electro optical modulator and a 80 MHz reference, for accurate analysis on a real interferometer. The results provide strategies for increased phase resolution for integrating the phase camera technology into the Einstein Telescope's alignment control system.

The thesis starts by an introduction of gravitational waves in section 2, and how they can be measured. It is discussed which technologies are used in existing second generation gravitational wave observatories like LIGO and which technological advancements third-generational gravitational wave observatories incorporate to reduce the different kinds of noise, which are introduced. Then it is discussed how the fundamental Gaussian beam is distorted through misalignment, which excites Higher-order modes in section 3. A way to measure this are phase cameras. A fiber-array based phase camera offers a way to resolve this in 2D with a higher speed than single-fiber phase cameras. The phase camera and the setup at ECAP is explained in section 4. The effect of synchronization on the timing is analyzed in section 7. Different calibration methods of the TARGET ASICs and their influence on the phase resolution are discussed in section 9. The implementation of the 500 MSa/s sampling rate, the timing optimization for this sampling rate and the influence on the phase resolution are discussed in section 10.

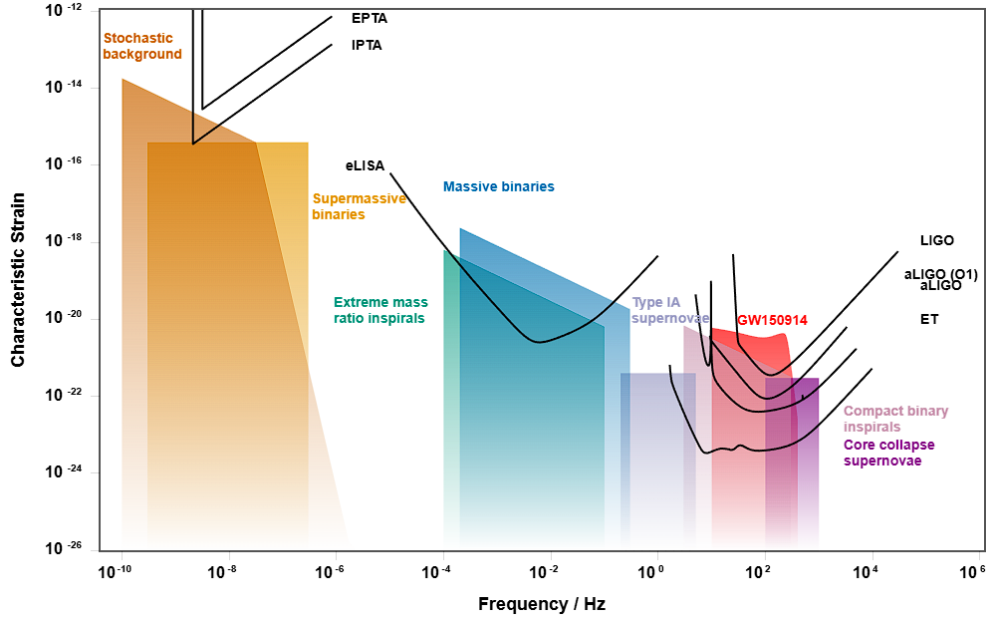


Figure 1: Characteristic strain of different sources of gravitational waves. Different sources are color coded. The sensitivities of different ground and space as well as pulsar timing based detectors are shown in black. Image taken from [22] adapted from [19]

2 Gravitational waves

This chapter introduces gravitational waves and the detection principle, outlines the technologies used in the Einstein Telescope, for which the phase-camera presented later is being developed.

Gravitational waves are ripples in space-time generated by the acceleration of massive, asymmetric objects such as merging black holes, binary neutron stars, supernovae or stochastic background. The waves travel through space at the speed of light carrying unaltered information about their source and the nature of gravity, since their interaction with matter is very weak [17]. This allows the waves to travel cosmic distances without absorption, although their amplitude is dissipating with distance, which means the signal is very faint upon reaching earth [18].

The initial gravitational wave detected by LIGO was generated by two colliding black holes 1.3 billion light years away with masses of $29M_{\odot}$ and $36M_{\odot}$ [19]. By the time the gravitational wave reached earth it caused a spacetime strain (h) of approximately 10^{-21} [20], resulting in a path length change of $4 \cdot 10^{-18}$ m [21], which is three orders of magnitude smaller than the nucleus of an atom cite. The different strains of different sources are depicted in Figure 1. Stochastic background and supermassive binaries produce the largest strain, while core collapse supernovae and compact binary inspirals create the smallest strain of approximately 10^{-22} . The detection of these strains requires a high sensitivity. The Einstein Telescope aims to extend this detection capability to higher sensitivities and extend the detectable frequency band.

To visualize these strains in spacetime caused by gravitational waves, the effect on a ring of particles is shown in Figure 2. The h_+ , $h_{\times}$ are the so-called plus and cross-polarizations in which a gravitational wave can travel [23]. When a gravitational wave propagates

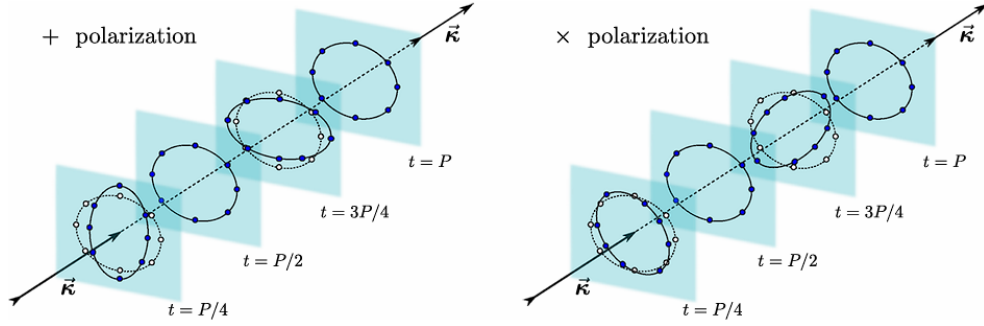


Figure 2: Schematic deformation of a ring of particles on the trajectory $\vec{\kappa}$ of a gravitational wave in the plus "+" and cross "times" polarizations. The wave elongates the ring in one direction while compressing in the direction perpendicular to it. The continuous lines and the dark filled dots show the positions of the particles at different times, while the dashed lines and the open dots show the unperturbed position. P is the period of the wave. Adapted from [6]

along a trajectory κ perpendicular to the ring's plane, it induces a characteristic quadrupolar deformation. The particles alternately oscillate along perpendicular axes in the plane orthogonal to the direction of wave propagation, stretching in one direction while compressing in the other [6].

The gravitational waves are quadrupolar. This means the gravitational wave is described by a symmetric, purely spatial metric perturbation that is both transverse to the direction of travel, which means the forces are only asserted perpendicular to their direction of travel and the deformations cause no overall expansion or contraction [24].

The primary challenge in gravitational wave detection is isolating the differential strain h , caused by a gravitational wave from the background of instrumental and environmental noise. Due to the quadrupolar nature of gravitational waves which causes a Differential change in perpendicular directions, an interferometer can be used to measure gravitational waves.

2.1 Interferometric Detection Principle

In an interferometer, a gravitational wave passing is represented by a periodic change in the time light takes to travel along the arms [25]. Because the wave is quadrupolar, it induces a differential change in the arm-length between the two orthogonal arms, creating a time and therefore phase shift of the two laser beams. The oscillating strain a gravitational waves causes, can be measured with a Michelson interferometer as shown in Figure 3. For visualization this interferometer is perfectly aligned to the deformation from the gravitational, but in general any interferometer in any direction can measure a passing gravitational wave, since only the differential arm length change is measured.

2.1.1 Interferometers

In an interferometer two sources of light are combined to create an interference pattern. The basic setup of a Michelson interferometer can be seen in Figure 4. A laser beam from a single source is split via a beam splitter, two mirrors reflect the beams back

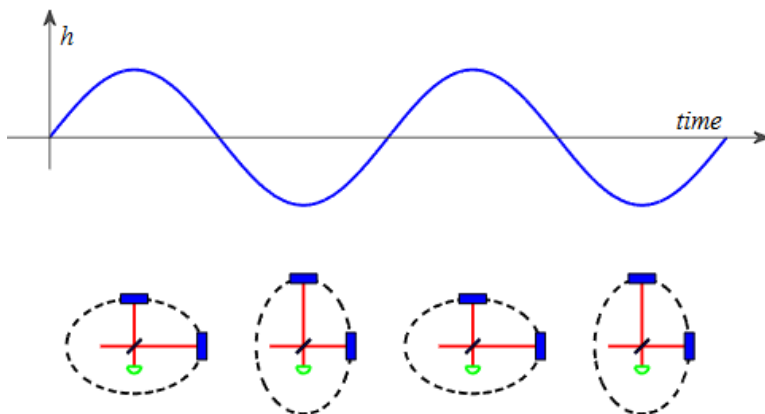


Figure 3: Gravitational wave strain h over time, with effect on a ring of particles (black). This strain can be measured with a Michelson interferometer (red). The length oscillations modulate the phase shifts accrued by the light in each arm, which are observed as light interference patterns at the photo detector (green). This depicts one of the linear polarization modes of the gravitational wave. Reproduced from [26]. Courtesy Caltech/MIT/LIGO Laboratory.

to the beam splitter where the two beams are recombined and then detected by a photo detector [27]. When the beams combine they interfere with each other, which means the amplitudes of the two beams are added at each point in time which forms an interference pattern. When the two beams constructively interfere, the amplitude rises, if they destructively interfere the amplitude falls. The interference pattern of two beams of laser-light depends on how well-aligned the light waves are when they are combined and contains various information from the beams path [27].

In gravitational-wave observatories the basic configuration of the Michelson interferometer is expanded to meet extreme sensitivity requirements. A vacuum system ensures the light travel is not diffracted in its path. Optical Fabry-Pérot cavities are integrated into each arm. These cavities act as resonators, forcing the laser light to bounce between the input and end mirrors multiple times, increasing the effective optical path (e.g., from 4 km to 1200 km for LIGO). This drastically increases the differential change in distance the light travels and therefore the phase shift induced by a gravitational wave for a given physical displacement, making the detection of a strain with amplitudes of $h \approx 10^{-21}$ possible. It also builds up the intensity of the beam, increasing the sensitivity of the detector, by increasing the amount of photons in the arms, which reduces quantum shot noise [7],[15].

The detector is operated at a dark fringe, where the arms are balanced so that the recombined light is destructively interfered at the signal-extraction (dark) port. Gravitational wave detectors reach their maximum sensitivity operating at dark fringe, so most of the light is reflected back towards the laser, with the so-called bright port [28]. A passing gravitational wave breaks this balance, causing the arms to oscillate, which results in a beating interference pattern carrying the gravitational-wave signal. The laser power, circulating into the detector is increased by introducing an additional mirror, which is called Power Recycling Mirror (PRM) at the bright port [28]. The setup of the PRM is shown in Figure 6. The Input Mode Cleaner (IPM) is a cavity,

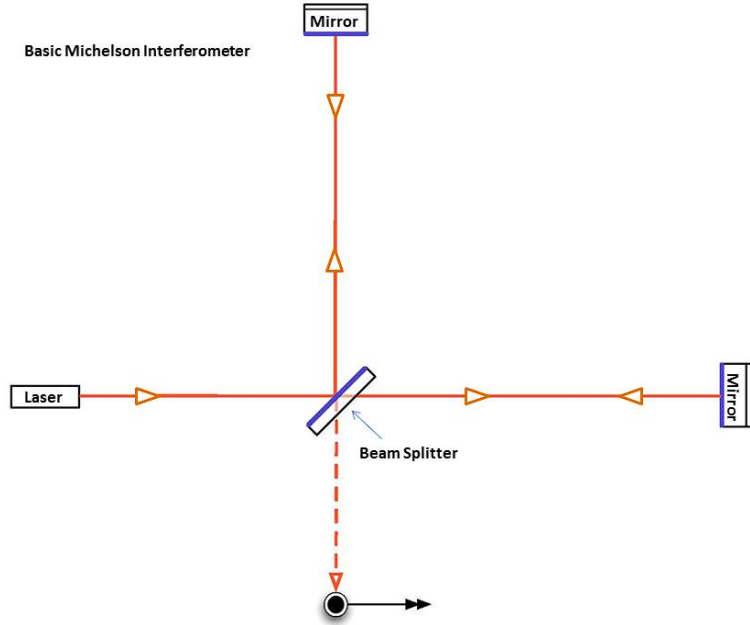


Figure 4: Basic schematic of a Michelson interferometer: The laser light is split with a beam splitter then travels along each arm. At the end of the arms it gets reflected, at the beam splitter the two beams are recombined and then the beam is detected with a photo detector. Image taken from [27]. Courtesy Caltech/MIT/LIGO Laboratory.

needed to improve the quality of the laser beam injected into the interferometer. A quantum noise reduction system is used at the signal readout to reduce the noises due to the quantum nature of light [29]. The output mode cleaner is a suspended glass optical cavity, for filtering out different frequencies, in the output beam of the interferometer [30]. The signal recycling mirror (SR) is an additional mirror placed at the dark port, which sends the signal back into the arm cavities. The optical system of the signal recycling cavity (from the dark port to the SR) and the arm cavities forms a composite resonant cavity, whose eigenfrequencies and qualities can be controlled by the position and reflectivity of the SR mirror. Near its resonances, the device can gain sensitivity. Detuning it allows the interferometer to work either in broadband or in narrowband configurations, and improving the observation of specific astrophysical gravitational wave sources [31].

Existing generation gravitational wave detectors already provided proof of gravitational waves and analyzed many sources. However, it is likely that the existing detectors will be operating until the end of the next decade. By then a complete overhaul of the infrastructure would be necessary to increase their sensitivities [14, p.20],[13, p.1]. The infrastructures are expected to have reached physical limitations in durability and performance, and won't allow further significant sensitivity improvements. A new third generation of Earth based detectors will be needed [13, p.1].

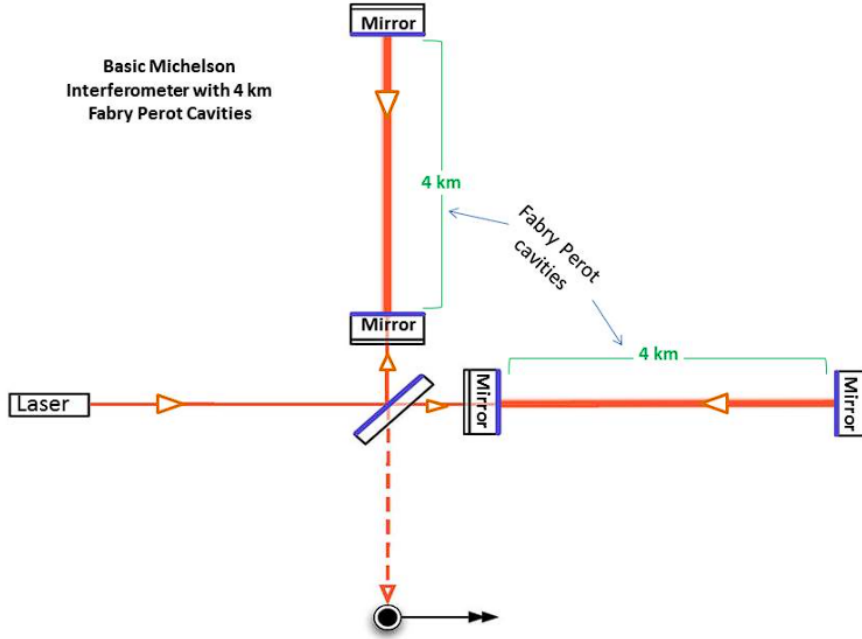


Figure 5: Basic Michelson interferometer with Fabry Perot cavities. Light gets reflected from each mirror at the end of the arms multiple times, which increases the optical path length [7]. Courtesy Caltech/MIT/LIGO Laboratory.

2.2 Einstein Telescope

This is the goal of the Einstein Telescope (ET), which is a third generation gravitational wave observatory, planned to start its activity in 2035 [13, p.1]. Gravitational wave detection opened a new window to the Universe. ET will begin expanding this window to unknown regions of the universe and examine formerly undetectable sources of gravitation waves, which will contribute significantly to the understanding of the early universe, the origin of heavy elements and the fundamental nature of gravity [13],[33]. It aims to surpass the limits of previous detectors. It will improve the sensitivity by an order of magnitude with respect to the design sensitivity of Advanced Virgo and LIGO and extend the observation band towards lower frequencies, down to about 3 Hz compared to about 10 Hz for Virgo. This will enable ET to detect binary black hole coalescences up to cosmological distances with a redshift of $z \approx 20$ and beyond. This will reach the dark era of the universe, preceeding the birth of the first stars. It will extend the region of black hole masses compared to that of second generation detectors, providing the first clear evidence of their existence. ET will detect the coalescence of binary neutron stars up to $z \approx 2 - 3$, which covers the vast majority of neutron stars. It could detect several new astrophysical sources of gravitational waves, such as signals emitted during core collapse supernovae, signals from isolated rotating neutron stars, or burst signals from neutron stars. ET will test several dark matter candidate, explore the nature of dark energy and search for stochastic background of gravitational waves [13, p.3-5].

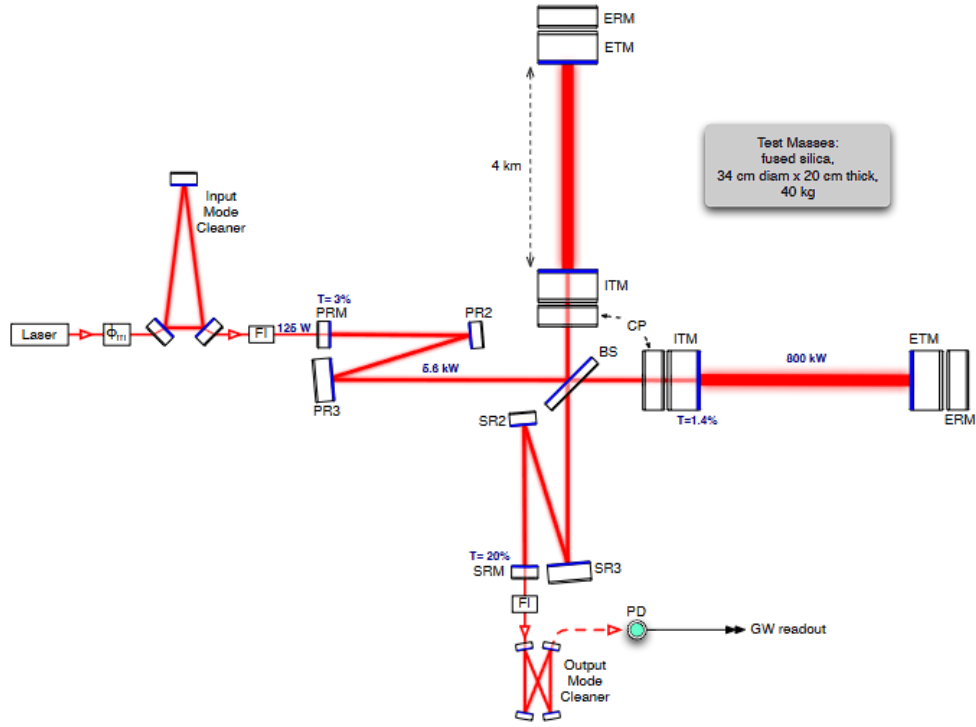


Figure 6: Advanced LIGO interferometer configuration: end test mass (ETM), input test mass (ITM), beamsplitter 50/50 (BS), end reaction mass (ERM), power recycling mirror (PRM), signal recycling mirror (SRM), power recycling cavity mirror 2/3 (PR2/3), signal recycling cavity mirror 2/3 (SR2/3), Faraday isolator (FI), phase modulation (ϕ m); photodetector (PD). The power levels shown correspond to full-power operation; the interferometers can also be operated at much lower powers with good strain sensitivity. Reproduced from [32]. Courtesy Caltech/MIT/LIGO Laboratory.

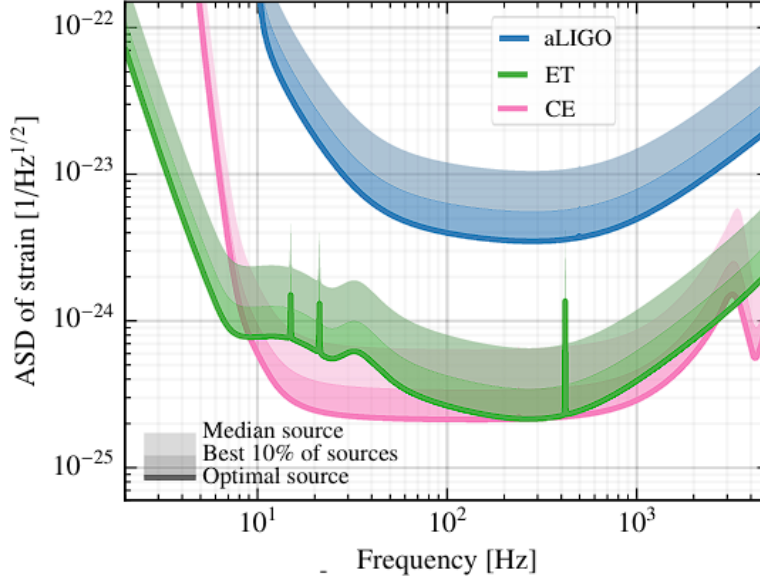


Figure 7: Design sensitivities of the gravitational wave detectors, advanced LIGO (aLIGO), Einstein Telescope(ET) and Cosmic Explorer (CE), a proposed next-generation ground-based gravitational wave observatory in the USA [34]. The various color shades represent different source locations and orientations. Reproduced from [14]

2.2.1 Design

ET will be based on the same basic concept, demonstrated by LIGO and Virgo. A modified Michelson interferometer, with Fabry-Pérot cavities in the arms and the techniques of power and signal recycling. The target sensitivity, in particular at low frequencies, is based on several technology innovations [13, p.1]. The sensitivities of the second and third generation gravitational wave detectors are shown in Figure 7.

Specifically, the ET detectors will have 10 km long arms, to increase the signal produced by the gravitational waves, compared to the 4 km long arms of LIGO [13, p.2]. The Einstein Telescope will feature two interferometers in each arm regardless of the configuration geometry. One will be optimized for high frequencies (HF) from 30 Hz to 10 kHz and one for low frequencies (LF) from 3 Hz to 30 Hz. Each interferometer is a Michelson interferometer with Fabry-Perot cavities with power and signal recycling cavities for arm cavity power enhancement and shaping the signal response. The lasers will operate at 1064 nm for the HF and at 1550 nm for LF. A schematic of the layout is shown in Figure 8 [35]. This separates the challenges related to the use of high power stored in the arms such as thermal and radiation pressure effects, from those related to achieve the targeted low-frequency sensitivity, which is limited by Brownian and quantum noise. The low-frequency detector will operate at a temperature of 10 K to 20 K and the high-frequency detector will operate at room temperature. To reduce the impact of seismic noise and gravity gradient noise induced by seismic waves and compression waves of the surrounding air, ET will be built underground. This also allows the extended bandwidth down to ≈ 2 Hz [13, p.3]. ET will incorporate the technologies from Advanced LIGO and Virgo detectors, like the ultra-sensitive optical interferometry, complex active and passive seismic isolation systems and injection of

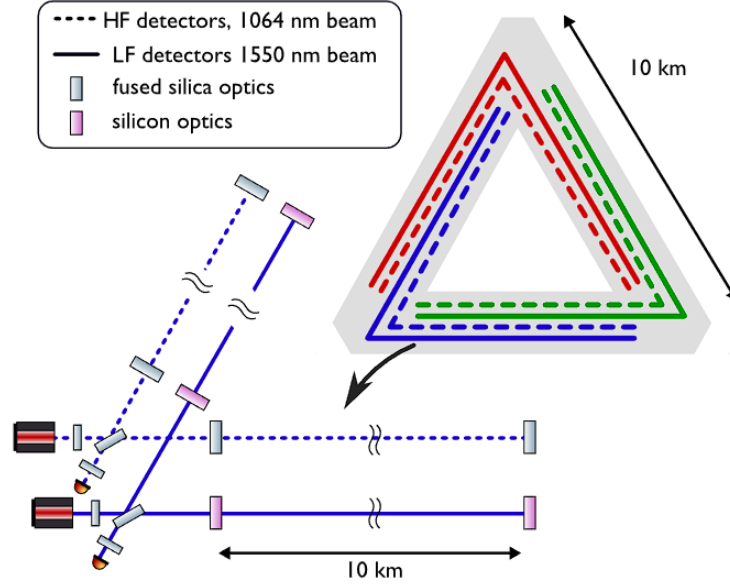


Figure 8: Diagrams for the Einstein Telescope layout with its triangle geometry in the top right corner and a schematic of the dual interferometer geometry for each detector. Each interferometer is based on a Michelson interferometer with Fabry-Perot cavities in the arms and recycling mirrors. Reproduced from [35, p.2]

non-classical squeezed light, as well as from planned Advanced Detectors, like cryogenic mirrors and frequency-dependent squeezing [13, p.2].

There are different configuration plans for the shape of the telescope. The original plan was to build a triangular detector with 10 km long sides and nested detectors, which means in each corner would be a laser that is split into the adjacent arms. Another possibility of two L-shaped detectors is examined, where the two detectors would be build at two different locations with a large distance between them [36]. A schematic of the two configurations can be seen in Figure 9. Both have advantages and disadvantages, which are not further discussed here. To achieve the increased sensitivity required for the ET, the noise sources that limited second generation gravitational wave detectors must be addressed. These range from fundamental quantum effects to technical instabilities, which demand high requirements on the optical configuration and control systems of the interferometer.

2.3 Noise in a gravitational wave detector

Different kinds of noises reduce the sensitivity of gravitational wave detectors. In Figure 11 the full noise spectrum of LIGO Hanford is shown. The detector noise in the frequency domain is measured in DARM, which is short for differential arm length [37, p.3].

Quantum noise Fluctuations of the vacuum electric field at the readout port defines a fundamental limit to the interferometer sensitivity. This quantum noise appears as shot noise and quantum radiation pressure noise. The uncertainty due to statistical

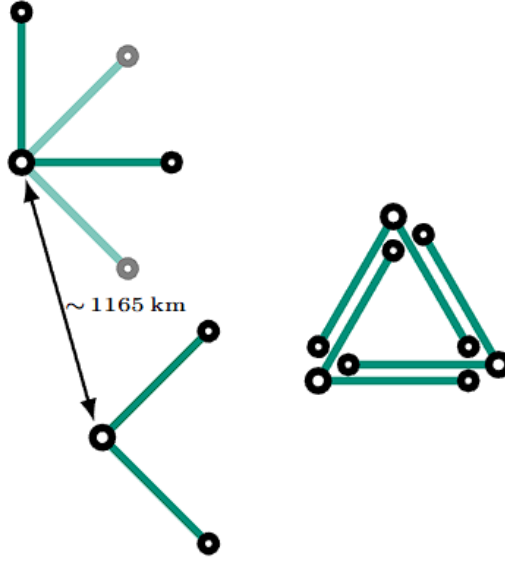


Figure 9: Schematic of different configuration designs that are being considered for the ET: two very distant L-shaped detectors (parallel, or 45 deg), or a triangle of three nested detectors. Reproduced from [36]

fluctuations in the arrival time of photons at the interferometer output results in photon-counting noise or shot-noise [15, p.7]. The shot-noise can be decreased by increasing the amount photons or laser power, which is achieved with the arm cavities and the PRM. Shot noise will be improved by the use of squeezed vacuum injected through the dark port of the interferometer. Injecting vacuum squeezed in the phase quadrature reduces the power fluctuations seen by the dark port photodiodes, which lowers the shot noise floor [15, p.7]. Increasing laser power introduces a second noise source. Radiation pressure noise is displacement noise and arises from amplitude fluctuations. Shot noise dominates the high-frequency region of the spectrum while quantum radiation pressure is a low-frequency noise, but doesn't dominate the low-frequency region. There are other non-quantum mechanical noise factors [15]. At low frequencies there is seismic noise, from ground motion by seismic and human activity. The underground location of ET reduces seismic noise [13, p.2]. The core optics must be suspended on multi-stage pendulum isolation systems to dampen these movements [15, p.7]. This suspension is shown in Figure 10[38].

Thermal noise caused by thermal motion of molecules in the test masses reflective coating results in displacement noise[15], dependent on the temperature of the optics [40]. This dominates the noise spectrum between 40 Hz to 100 Hz. ET will incorporate cryogenic mirrors to reduce Brownian noise [13]. Scattered light noise comes from light that scatters from imperfections of the optics, like a test mass, hitting surfaces with motion relative to the optic, like the vacuum chamber walls. When this scattered light reflects back into the beam and the distance to the moving surface is changing due to vibrations, this causes fluctuations in the beam amplitude [15, p.11].

There are many other sources of noise depicted in Figure 11, which are not discussed in detail here, but still influence the measurements at gravitational wave detector facilities

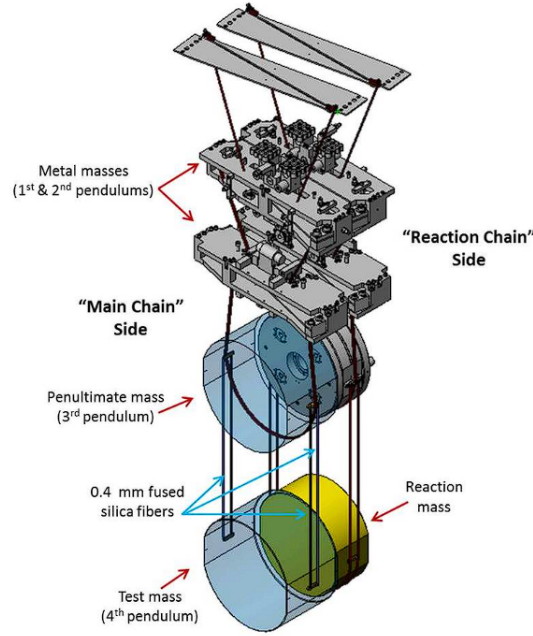


Figure 10: LIGO’s mirrors, called test-masses, are suspended from a four-stage pendulum by glass fibers, acting as a passive stabilization against disturbances. Reproduced from [39]

like LIGO and ET so their sensitivity is governed by that noise.

The discussion of the noise so far only considered good alignment conditions of the mirrors with respect to the optical axes. If left free in their angular degrees of freedom, the mirrors will be subject to motions, due to the residual seismic noise at low frequency. There are different control systems, like local sensors and a global automatic alignment system to control the alignment [16, ch.4]. To understand the effect alignment has on the laser beam, first gaussian beams need to be introduced.

3 Gaussian beam

This chapter explains the properties of a Gaussian beam with a derivation of the solution of the paraxial wave equation. Then the influence of misalignment and higher-order modes is discussed, which is measured with the fiber-array based phase camera.

In a general case the propagation of light inside an optical system is described by the Maxwell equations and by the corresponding wave equation [16, ch.4].

$$(\Delta + k^2) \mathcal{E}_k(x, y, z) = 0 \quad (1)$$

In an interferometer, laser propagation is often described in the paraxial approximation. This assumes the optical field travels predominantly along the z-axis and varies only slowly in that direction [16, ch.4]. Writing the field as

$$\mathcal{E}(x, y, z) = e^{ikz} E(x, y, z) \quad (2)$$

By assuming the envelope changes are small over one wavelength, the wave equation reduces to the paraxial diffraction equation Equation 3, which is appropriate for

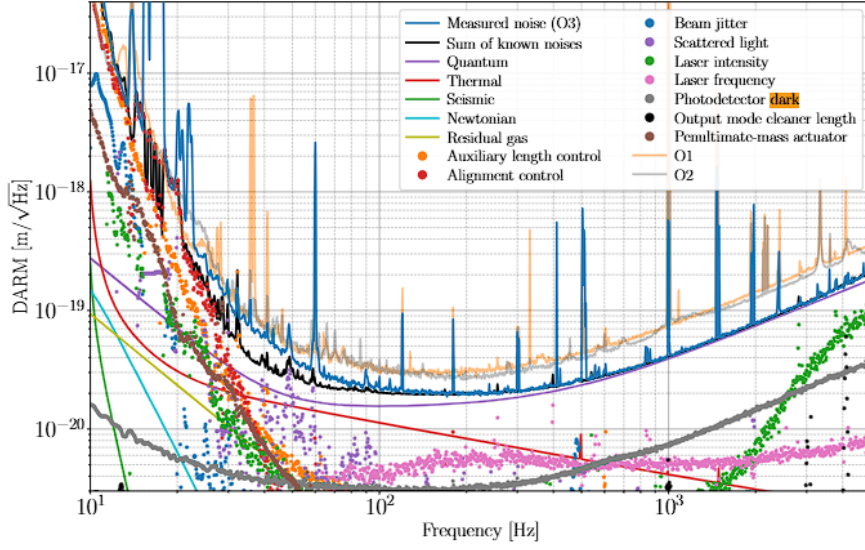


Figure 11: Noise spectrum for LIGO. Reproduced from [15]

nearly monochromatic beams that remain in a small area confined around the axis of propagation [16, ch.4].

$$(2ik\partial_z + \Delta_{trans}) E = 0 \quad (3)$$

Under cylindrical symmetry the simplest solution is the fundamental Gaussian beam [16, ch.4],

$$\Psi(r, z) = \Psi_0 \frac{w_0}{w(z)} \cdot e^{-\left(\frac{r}{w(z)}\right)^2} \cdot e^{-ik\frac{r^2}{2R(z)}} \cdot e^{i(kz - \zeta(z))} \quad (4)$$

with the following relations for the Rayleigh length z_R , the beam width $w(z)$, the beam curvature $R(z)$ and the Guoy phase $\zeta(z)$ [16, ch.4]. The Rayleigh length is the distance from the beam waist at which the cross section of the beam doubles, the beam radius has increased to $\sqrt{2}w_0$ and R is minimal [41],[42].

$$z_R = \frac{kw_0^2}{2}, \quad w(z) = w_0 \sqrt{1 + \frac{z^2}{z_R^2}}, \quad (5)$$

$$R(z) = z \left(1 + \frac{z_R^2}{z^2}\right), \quad \zeta(z) = \arctan(z/z_R) \quad (6)$$

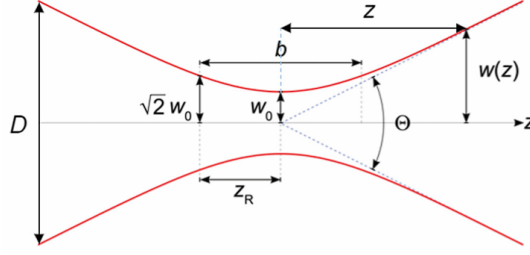


Figure 12: Beam parameters: waist diameter ω_0 , beam divergence half angle θ , Rayleigh length Z_R . Image taken from [43, p.4]

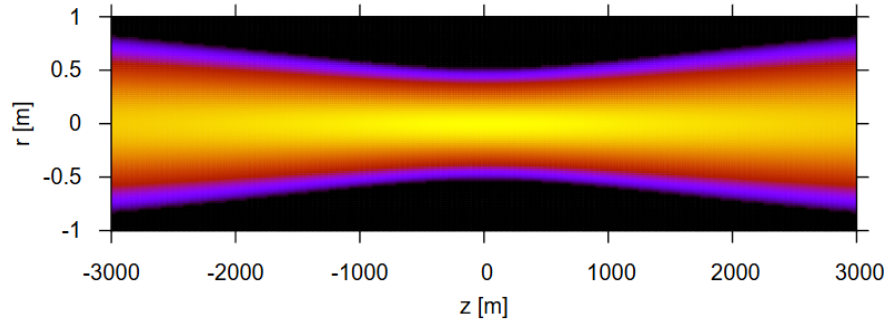


Figure 13: Intensity map of a Gaussian beam with logarithmic color scale. Reproduced taken from [16, p.53]

The transverse intensity shape is a Gaussian function for every z . The beam waist at $z = 0$ is the point, where the radius is the smallest and increases for larger z according to the function $w(z)$, which can be seen in Figure 13. The wavefronts of this beam are parabolic, described by $R(z)$. A propagation introduces the Guoy phase shift shown in Equation 7 [16, ch.4].

$$\phi_G = -\arctan \frac{z}{b} \quad (7)$$

The measurable output of an interferometer is the intensity at the detector, which depends on the relative phase of the interfering waves through the interference term in Equation 8 [44, p.1].

$$I_{out} \propto |\Psi_1 + \Psi_2|^2 = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos(\Delta\Phi) \quad (8)$$

3.1 Transverse electro-magnetic modes

Starting from the fundamental solution of Equation 4 it is possible to describe more general field shapes by the introduction of Hermite-Gaussian beams. These extend the Gaussian beam, by allowing different transverse structures in x and y . The Hermite-Gaussian beams are also called Transverse Electromagnetic modes (TEM) and are labeled with two integer indexes n and m . The normalization coefficient N_{mn} is chosen to make this set of TEM orthonormal, and the $H_n(t)$ are the Hermite polynomials. The first order Hermite polynomials are shown in Table 1 [16, ch.4].

$$TEM_{mn}(x, y, z) = N_{mn}(z) e^{ikz} H_m \left(\frac{\sqrt{2}x}{w(z)} \right) H_n \left(\frac{\sqrt{2}y}{w(z)} \right) e^{-i(n+m+1) \arctan(z/b)} e^{ik \frac{x^2+y^2}{2R(z)}} e^{-\frac{x^2+y^2}{w^2(z)}} \quad (9)$$

Order	Polynomial
0	1
1	$2x$
2	$4x^2 - 2$
3	$8x^3 - 12x$

Table 1: Hermite polynomials of lowest order [16]

For a fixed value of the beam size at the waist w_0 , these solutions form a complete, orthonormal set, meaning that every beam can be expressed as a linear combination of these functions at fixed z [16, ch.4].

The transverse intensity pattern of the first TEM modes are shown in Figure 14.

These TEMs introduce a new Guoy phase shift of

$$\phi_G = -(n + m + 1) \cdot \arctan \frac{z}{b} \quad (10)$$

[46]. This Guoy phase shift includes the order of modes m, n . For the fundamental Gaussian beam, which is the TEM_{00} mode, the general solution for higher order modes Equation 9 reduces to the fundamental Gaussian beam solution.

3.2 Resonant cavities

In general, when a TEM mode with a defined waist size and position is reflected from an arbitrary mirror, the shape of the beam will change, which means it is not possible to describe the beam as a single TEM mode with the same beam size and position. Only a fraction of the reflected beam will be in the same mode, which reduces the effective reflectivity. In the Fabry-Pérot cavities, consisting of two mirrors the laser beam should stay in same mode as the input mode, which is the fundamental mode. This is achieved with a different shape of the mirrors in the cavity. If the shape of the mirror follows the wavefront of the incident beam, which is the surface of constant phase, the reflected beam will be in the same mode as the incident one. The optimal shape for a resonant cavity is curved. Typically the input mirror of a resonant cavity is flat and the end mirror is curved, which needs the input mirror to be placed at the beam waist, where the wavefront is planar. The wavefront curvature at the end mirror is defined by $R(L)$ from Equation 6 and the length of the cavity L . A resonant mode only exists if the radius of curvature of the end mirror is larger than the length of cavity [16]. In the resonant condition the phase difference $\Delta\Phi$, which the beam experiences over one cycle must be equal to a multiple N of 2π . With the phase from Equation 9, the full $\Delta\Phi$ consists of the plane wave phase term $2kL$ with the length of the cavity L and the wave vector $k = \frac{2\pi}{\lambda}$ from propagation, the phase shift induced by the mirror $+\pi$ and the Guoy phase from Equation 10 [16, ch.4][47].

$$\Delta\Phi = 2kL + \pi - 2(n + m + 1) \cdot \arctan \frac{L}{b} = 2\pi N \quad (11)$$

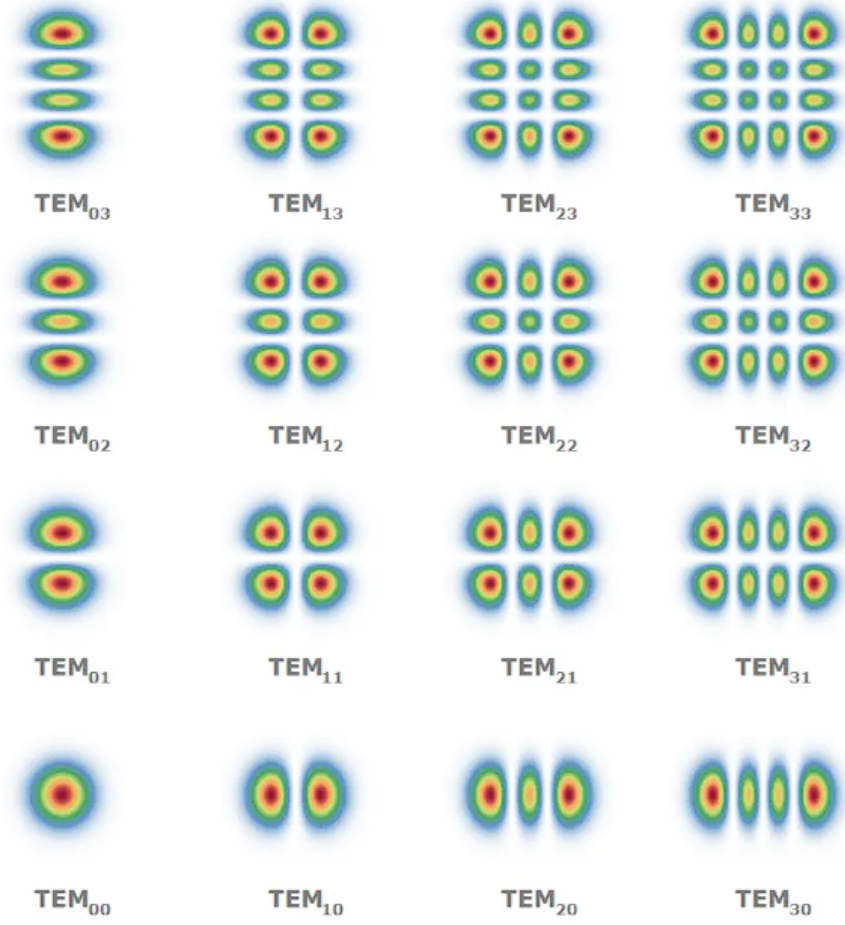


Figure 14: Intensity profiles of Hermite-Gaussian modes, starting with the fundamental gaussian mode in the lower left corner and expanding to TEM_{33} in the upper right corner. TEM stands for transverse electromagnetic wave. Reproduced from [45]

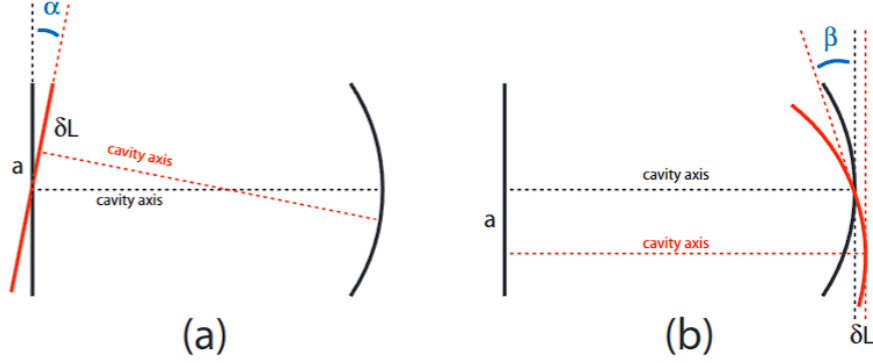


Figure 15: Schematic misalignment of the two mirrors in a cavity and their influence on the optical axis. Image taken from [16, ch.4]

This shows, that if the $TEM_{0,0}$ mode is resonant, all other modes are not, except in the case of degeneracy [16].

3.3 Cavity response to misalignment

The optical axis of a resonant cavity, is the only line perpendicular to both mirrors. A tilt of one mirror changes the optical axis of the cavity, visualized in Figure 15. Through a coordinate system change, one can derive the effect of input or end mirror misalignment in the cavity optical axis reference system, as the creation of additional transverse modes proportional to the tilts. This is described by

$$\begin{aligned}
 TEM_{0,0}^{IN} = & TEM_{0,0}^{CAV} + \alpha \left[\frac{R_C - L}{w_0} + i \frac{k}{2w_0} \right] TEM_{1,0}^{CAV} \\
 & + \beta \frac{R_C}{w_0} TEM_{1,0}^{CAV}
 \end{aligned} \tag{12}$$

where α and β are the angles of misalignment of front and end mirror respectively and R_C is the curvature of the end mirror [16, ch.4]. This can also be generalized that substrate refractive index inhomogeneities and thermal effects from optical absorption in test masses distort the beam from the ideal Gauss profile and excite higher-order modes [48]. Therefore, the mode content needs to be measured to accurately control the length and alignment of the optical cavities.

3.4 Alignment sensing and control

Gravitational wave detectors therefore employ dedicated alignment sensing and control mechanisms to monitor the beam position and angular stability of the arm cavities. Multiple sidebands for length and alignment sensing and control of key optical cavities are used. These are frequency phase modulation sidebands at multiples of 10 MHz, modulated onto the main beam by the electro-optical modulator (EOM), via the Pound-Drever-Hall laser stabilization technique. A schematic of the principle of sidebands is shown in Figure 16. The beating between carrier and sidebands is filtered by different analog and digital electronics and fed back to the control system [15, p.3]. Several sidebands are used to control the different longitudinal and angular degrees of freedom

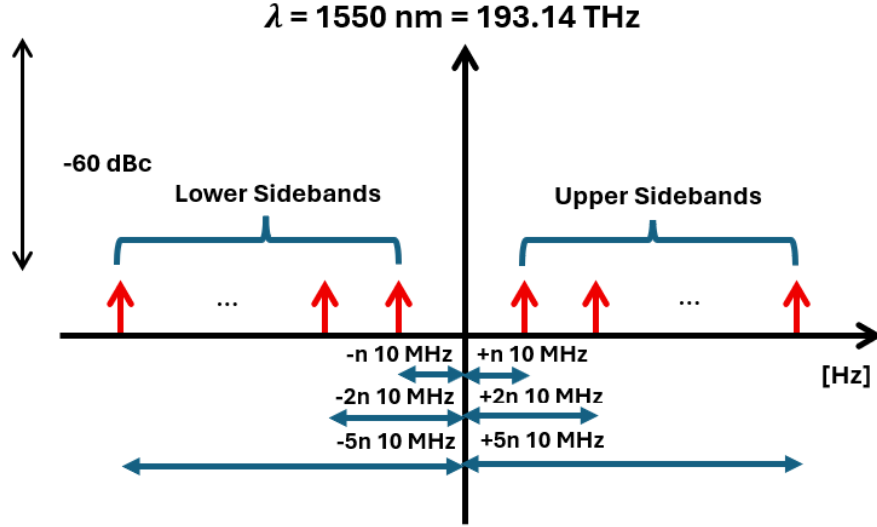


Figure 16: Sidebands modulated onto the carrier frequency with a shift of 10 MHz. Reproduced from [48]

[15, p.22]. 10 MHz sidebands are a practical modulation choice. The sidebands can be chosen to match available equipment but need to be large enough from the cavities resonance to provide an error signal, while still being within the photodetector bandwidth [49, p.10].

4 Phase camera

In this chapter the fiber-array based phase camera as a way to measure the mode-content is introduced. The method to calculate the phase from the laser intensity is described. The phase gives information about the mode-content since modes introduce the Guoy phase. This motivates the need for an increased phase resolution to resolve the modes more easily.

Next-generation detectors such as ET require simultaneous monitoring of the phase content of the carrier and multiple sidebands (± 10 MHz to 100 MHz). This information is needed to accurately determine the different longitudinal and angular degrees of freedom of the interferometer by analyzing the mode content [15].

The phase of the laser is measured by interfering the signal beam at the signal port with a reference beam, which is frequency shifted by 80 MHz using an acousto optical modulator (AOM), as shown in Figure 17. This is called heterodyne detection. Optical heterodyne detection is the mixing of a signal with a strong local oscillator wave to generate an electrical beat note at their difference frequency [50]. This frequency offset separates the different sidebands, so the different sidebands can be distinguished more easily [48]. Without the reference beam, sidebands such as ± 10 MHz are not separable, with the 80 MHz offset, they appear at frequencies of 70 MHz and 90 MHz relative to the carrier frequency, which is visualized in Figure 18. The interference between the reference and the carrier beam generates a beating signal, encoding the phase difference $\Delta\Phi$. This beating signal is represented by the equation Equation 13, derived from Equation 8. The intensity measured is proportional to a DC term, A , Φ and the

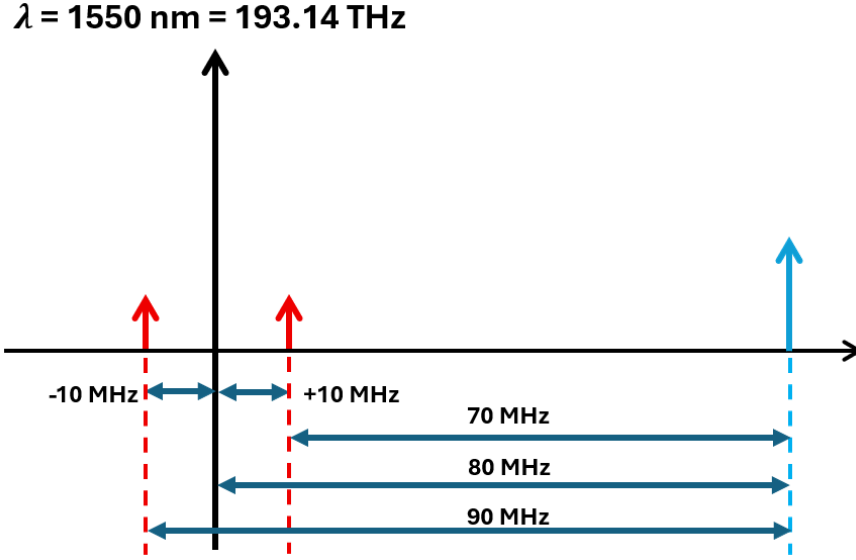


Figure 18: Optical heterodyne measurement with a 80 MHz reference beam creates sideband differentiation. Reproduced from [48].

frequency difference from the main beam Δf , which is 80 MHz for the carrier frequency and $80 \text{ MHz} \pm n \cdot 10 \text{ MHz}$ for the sidebands Equation 13.

$$I(t) = DC + 2E_{1550 \text{ nm}} E_{\text{ref}} \cos(2\pi \Delta f t + \Delta \phi) \quad (13)$$

where Δf is the frequency difference of the two beams, which is 80 MHz [48].

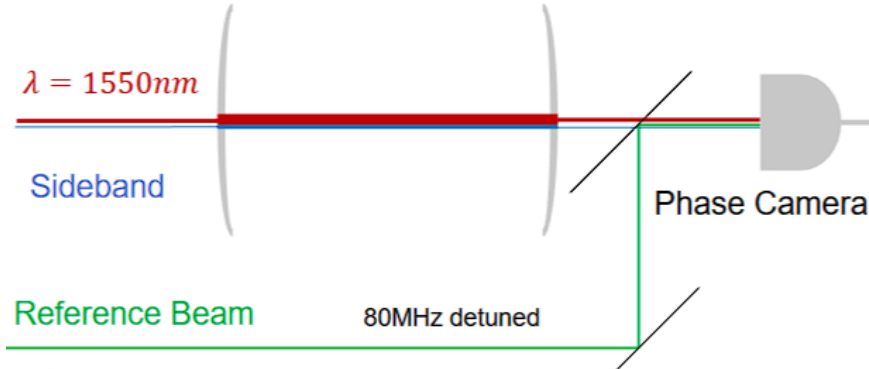


Figure 17: Schematic of optical heterodyne detection with a reference beam. Reproduced from [48].

4.1 Fiber array

The heterodyne beating is then sampled with a phase camera. Single-pixel phase cameras implemented in Virgo use a single pixel photo diode which is scanned across the beam with a Piezo mirror and generates a "100 x 100" pixel image each second. A novel 2D fiber array camera was developed to address the issue of sampling time. The

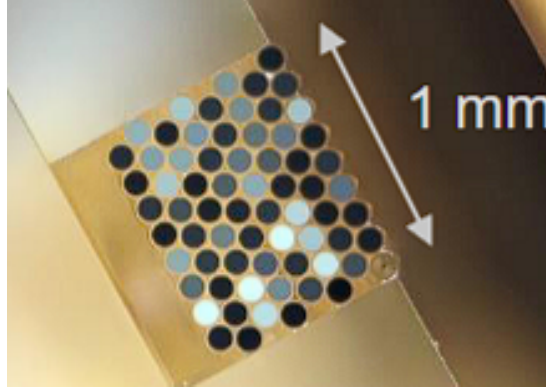


Figure 19: Fiberarray with 8 times 8 optical fibers. Image taken from [48]

laser is captured by an optical fiber array. The prototype optical fiber with an array of 8 times 8 optical fibers can be seen in Figure 19. This setup provides a 2D resolved image of the laser phase content which then can be analyzed to visualize the spatial phase content of the laser and therefore determine if the laser is perfectly aligned.

4.2 Laser setup

For future incorporation into laser interferometers like the Einstein Telescope, the capabilities of the fiber-array phase camera on a laser interferometer need to be tested. For that purpose there is a interferometer setup in the ET Laboratory at ECAP which can be seen in Figure 20. The input laser beam comes from the bottom right (red), it is then split with a beam splitter into the main beam (blue). The electro optical modulator (EOM) modulates the sidebands onto the main beam at frequencies ± 10 MHz. The reference beam (orange) goes to the acousto-optical modulator (AOM), which detunes it by 80 MHz. At another beam splitter they are recombined and then directed at the Fiber array. The setup of the phase camera was tested at the ET pathfinder in Maastricht [51, ch.7]. The ETpathfinder is a R&D infrastructure, located in Maastricht, for testing and prototyping innovative concepts and enabling technologies for the Einstein Telescope [52]. A 1D-fiber array was scanned across the beam to create a pseudo 2D image of the wavefront of the laser [51]. A 2D image of a TEM_{00} mode, so the fundamental Gaussian beam is shown in Figure 21 [51]. This proved the working principle of the fiber-array based phase camera. The fiber array phase camera can capture the wavefront spatially resolved in 2D. Since we are interested in the phase of the wavefront, the digital data must be demodulated. This is done via IQ-demodulation.

4.3 IQ-demodulation

To characterize the wavefront distortions and higher order mode content of the interferometer output, the complex amplitude A and phase Φ of the beating pattern captured by each pixel must be analyzed. This is a heterodyne beating, resulting from the interference of the output beam and the frequency-shifted reference beam.

To extract the phase of the signal, it is digitally demodulated. In a normal amplitude demodulation only the real part of a complex signal is calculated. This means only the amplitude is analyzed, the phase information is lost [53]. Therefore, we need another

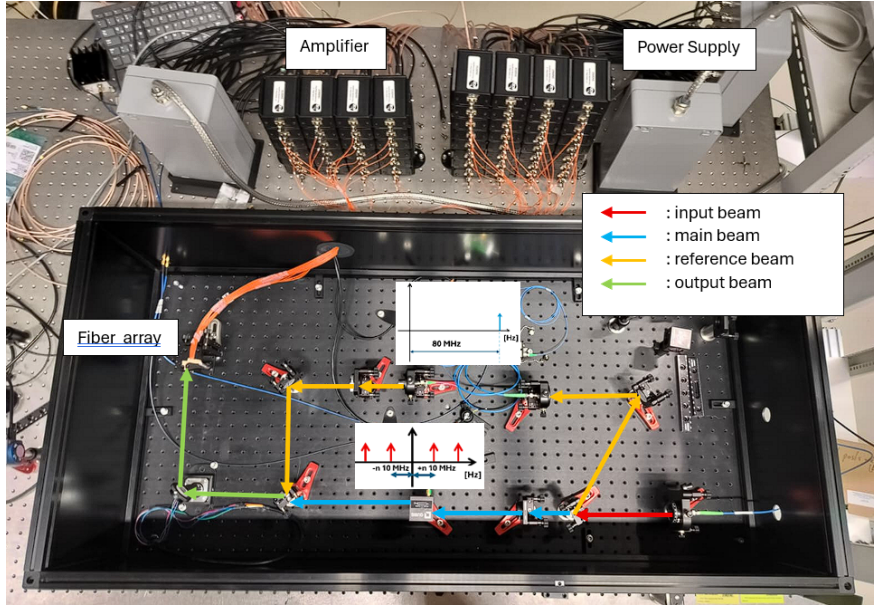


Figure 20: Laser setup in the ET Lab at ECAP

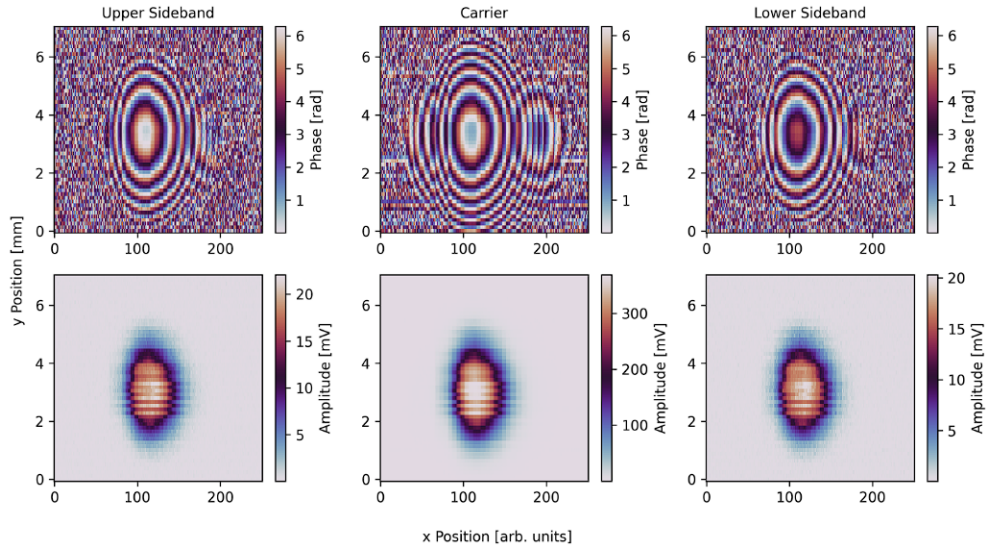


Figure 21: Pseudo 2D phase and amplitude images of beam configuration B for both sidebands and the carrier in the fundamental TEM₀₀ mode. The additional concentric structures, left and right in phase, are a Moiré pattern due to the ratio between the phase curvature of the wavefront and the pixel width in the x-direction. Reproduced from [51, fig. 48].

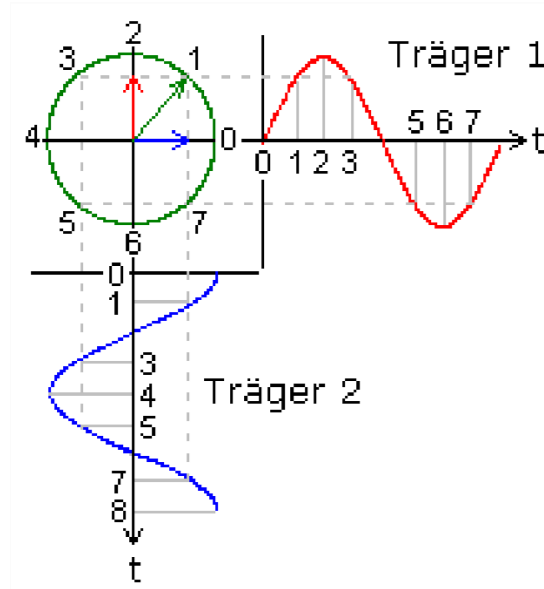


Figure 22: I and Q signals as parts of a complex waveform visualized on the complex plane. In red the I signal is shown as a sine, in blue the Q signal as a cosine, and their corresponding parts in the complex plane. Image taken from [56]

method called IQ demodulation. This shifts the beating from its frequency Δf to baseband, while preserving amplitude and phase information [54]. In this context baseband refers to a frequency range centered around 0 Hz. The frequency down conversion is performed by multiplying the 80 MHz beating with an complex sinusoid signal at the same frequency Δf [55].

$$x_{IQ}(t) = x_{80MHz}(t) \cdot \exp(-i2\pi\Delta f \cdot t) \quad (14)$$

IQ demodulation splits the incoming signal into two branches, the in-phase and quadrature signals. Since the sinusoid is complex it can be split into the "In-phase" signal I , which is the signal multiplied by $\cos(2\pi\Delta f \cdot t)$, and the "Quadrature" signal, which is the signal multiplied by $\sin(2\pi\Delta f \cdot t)$, which means the quadrature signal is 90° phase shifted to the in-phase signal [53]. By performing low-pass filtering to remove the high-frequency $2\Delta f$ components, the negative frequencies are removed. In our implementation, these components are obtained by integration, a windowed summation that acts as a low-pass filter to suppress noise outside our desired bandwidth [55]. The I,Q signals can be represented by the following equations [48].

$$I = A \cos(2\pi\Delta f \cdot t) \quad (15)$$

$$Q = A \sin(2\pi\Delta f \cdot t) \quad (16)$$

These components represent the projection of the signal onto the orthogonal axes of the complex plane, as illustrated in Figure 22 [56].

The phase angle can then be calculated with Equation 17 and the absolute amplitude of the signal via the Pythagorean theorem with Equation 18 [Tevpa].

$$\Phi = \arctan2(Q, I) \quad (17)$$

$$A = \sqrt{I^2 + Q^2} \quad (18)$$

The arctan2 function is used to ensure the phase is mapped correctly across $([-\pi, \pi))$, which is essential for full waveform reconstruction [57]. By performing this demodulation for all pixels in the fiber array in parallel, the full 2D wavefront geometry can be mapped, which reveals the beam's spatial mode content and optical distortions directly.

5 Measurement devices

5.1 TARGET ASICs

In this chapter the TARGET ASICs are introduced as a way to sample and digitize the analog input from multiple fibers from the fiber-array based phase camera for digital IQ-demodulation. These TARGET ASICs are characterized and optimized in this thesis.

The fiber array requires the simultaneous digitization of 64 or more spatial channels. A setup with individual traditional digitizer cards would be complex and expensive. We therefore use TeV Array Readout Electronics with gigasample per second (GSa/s) sample rate and Event Trigger (TARGET) application-specific integrated circuits (ASICs), which is a set of custom ASICs originally designed for the readout electronics of Small-Sized-Telescope (SST) of the Cherenkov Telescope Array (CTA) [58], to achieve compact, cost-effective, and high performance waveform digitization. They are designed for high-speed, multi-channel waveform capture, providing GSa/s sampling bandwidth for 16-channels simultaneously [48]. This makes them suitable to process the heterodyne beating from the fiber-array in parallel. Sampling and digitizing is done by the ASIC Cherenkov TARGET C (CTC), with a 16.384 sample deep storage buffer and the ability to digitize at arbitrary times. The companion trigger unit is the Cherenkov TARGET 5 Trigger Extension ASIC (CT5TEA). The CTC is suitable for continuous sampling without dead time and random access full waveform readout [58]. The multi-channel TARGET ASICs are very cost efficient in comparison to single channel digitizers [48].

5.2 Architecture of the ASICS

In the following the operational principle of the TARGET ASICs and their architecture seen in Figure 23 is described. The pair of ASICs is divided into the trigger unit (CT5TEA) and the analog sampling buffer with analog-to-digital converters (ADCs) for signal digitization (CTC) and digital-to-analog converters (DACs) for internal operation settings. The analog values are stored in CTC in a small Switched-Capacitor Array (SCA) of 32+32 cells. These cells are following the signal until they are disconnected sequentially with a 1 ns delay, which is generated by a temperature-stabilized internal time base. These cells are following the signal until they are disconnected sequentially with a 1 ns delay, which is generated by a temperature-stabilized internal time base. The SCA consists of two blocks of 32 cells, which allows for alternating sampling and readout to reduce dead time. The signal path is shown in Figure 23. The operational pedestal, which is an adjustable DC baseline per channel, is set by DACs in CT5TEA. The onboard FPGA, short for Field-Programmable Gate Array [59], manages the complex timing required for the SCA sampling, trigger-logic implementation, and Ethernet-based data transfer, serving as the bridge between the analog signal sampling and the digital

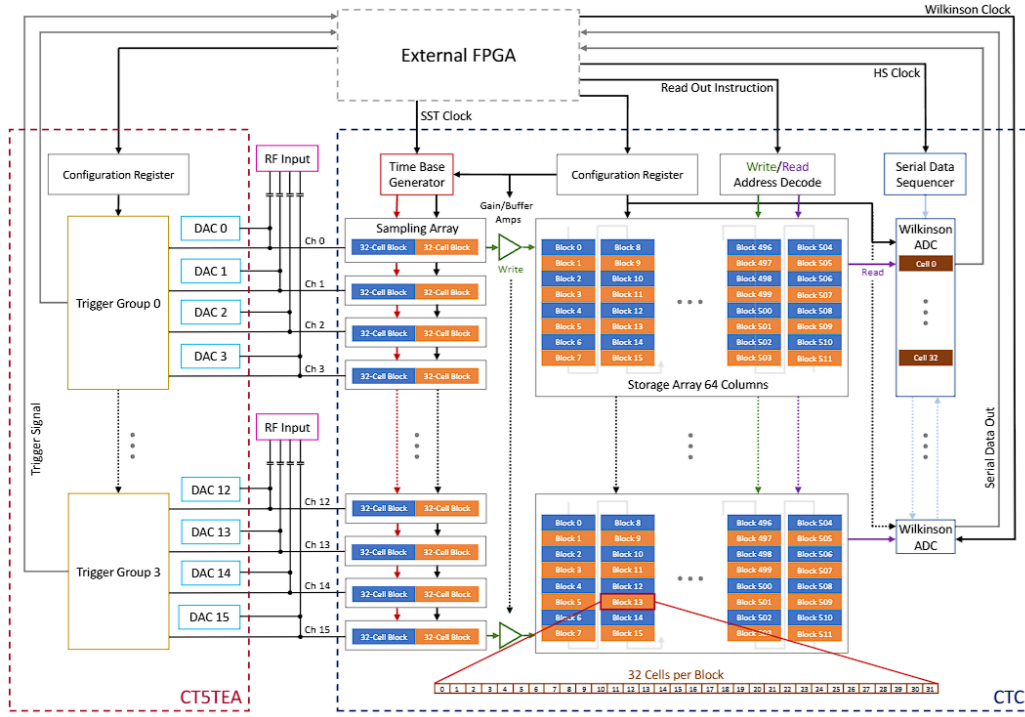


Figure 23: Operational schematic of the TARGET CT5TEA and CTC ASICs. Image taken from [58].

phase reconstruction software [58]. The readout signal is transmitted to the 16,384 sample deep ring buffer storage array. This consists of 512 blocks of 32 cells, so 16 μ s of data can be buffered at 1 GSa/s. The samples are constantly overwritten until a trigger signal signals data readout. The position in the buffer where the trigger occurred is blocked from being overwritten while the sampling continues for the rest of the storage buffer. A trigger is an instruction for digitization. For each trigger 448 samples are digitized. Each cell of one 32-cell block is digitized in parallel by Wilkinson ADCs. At the start of the digitization, a 12-bit Gray code counter is reset and counts up with the speed of the externally applied Wilkinson clock. The Wilkinson ADCs compare the stored cell voltage with a reference voltage ramp, which is generated by an external capacitor. When the ramp voltage gets larger than the storage cell voltage, the current value of the counter is written as the digitized value. It takes about 20 μ s to digitize a full block [58].

5.3 FPGA Firmware

Since the FPGA controls the TARGET ASICs, the firmware for the FPGA needs to be discussed. Changes to the firmware were made during the duration of this thesis to optimize the modules. The firmware is written in vhdl, which is short for Very High speed integrated circuit hardware description language. This is a description language, contrary to C++, which is a general-purpose programming language [60] for low-level programming. It is used to develop analog and digital circuits, which describes the behavior of a circuit on a higher level than the single electronic elements

[61]. Starting point is modeling the behavior of the circuit. The circuit is structured into function blocks, to receive structural descriptions. This process is repeated for different structural components until the circuit can be practically realized. The hierarchy is setup in multiple steps. First components are declared and then instantiated, where the connecting signals are mapped onto the ports [62]. The top-level entity of the firmware is `APT_Eval.vhdl`. It acts as the root module declaring primary input/output ports, instantiating sub components and connecting them via port mapping. This means lower-level modules receive signals through their port definitions. Internal signals, that are declared in the architecture route data between components like `HM_GLink`, which is responsible for managing the Ethernet connection from the board to the PC it is controlled by. In `APT_Eval` the module `TARGET_blk_tack_t7` is also instantiated. This is a top module for different sub modules, which are responsible for everything related to sampling and digitizing, as well as the the serialization of the data and the export and the FIFO module. This module manages the buffering logic that temporarily holds data between different parts of the design. There are other modules that are less relevant here, which control synchronization, communication or trigger and timing. The firmware versions used for measurement were `CTC_Eval_0xA0000017` and `CTC_Eval_0xA0000018` for the 1 GSa/s and 500 MSa/s sampling rates.

5.3.1 TARGET evaluation board

The TARGET evaluation board (Evalboard) is a prototyping board designed to test both CTC and CT5TEA and their ability to measure data from 16 optical input channels for phase measurements. This is our experimental test bench for the ET phase camera, testing the scalable solution for the spatial phase reconstruction for the Einstein Telescope’s alignment control system for reaching unprecedented sensitivity. The data is sampled, stored on the TARGET CTC chip and then digitized and send via Ethernet connection to the PC. The onboard FPGA A picture of the Evalboard on the upper site can be seen in Figure 24, with the FPGA connect and Ethernet connector, 16 MMCX input connectors on the left, four user- configurable IO ports, and Trigger input and output on the right [58].

5.4 Measurement with Evalboards

The Evalboards are controlled with a python interface to the software. The Software for operation of the TARGET Evalboards, is written in C++. The `targetdriver` module provides the low-level TARGET hardware driver that manages the connection to the TARGET modules via UPD client. In `targetio` module all Input Output operations are defined. This covers the the write/reading of the event files. The `targetcalib` module computes and applies calibration products, like pedestals and DC transfer functions, which are explained later. Then there is the Python front-end, which contains different scripts to connect with the Evalboards, measurements and analyses. It also contains different configuration files for the TARGET modules, calibration scripts and analysis functions and scripts.

The measurement is controlled with python files. First the `targetio` module and other python modules are imported and data paths are given. Some parameters are set, like the duration of measurement, warm-up time and multievent. The firmware is set to readout 448 samples per trigger, or 14 blocks times 32 samples/block. For sampling

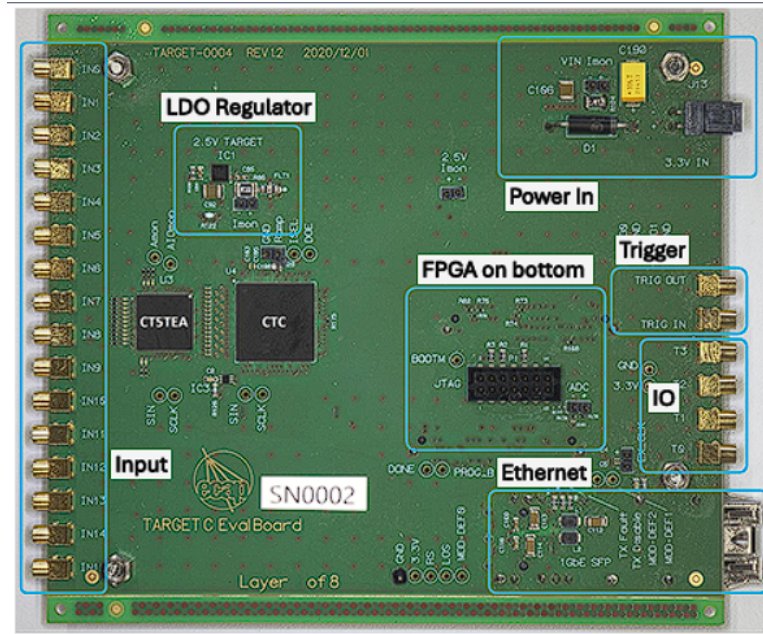


Figure 24: One of the two evaluation boards used to operate CTC and CT5TEA with the FPGA and Ethernet connector (SFP port) on the back side, 16 MMCX input connectors on the left, four user- configurable IO ports and Trigger input and output on the right. The board is supplied with a single 3.3V supply (Molex Nano-Fit connector top right). Reproduced from [58]

longer waveforms, a setting called multievent is employed. In this mode, a single hardware trigger pulse initiates a sequence of $N + 1$ consecutive acquisition events. These acquisition events form one trigger event for each physical trigger. While the physical trigger is only asserted once per cycle, the FPGA logic generates N additional triggers, each offset by a fixed interval of 448 samples. These acquisition events form one trigger event for each physical trigger. The trigger event is a continuous temporal window of 4032 samples for multievent $N = 8$ ($448 \text{ samples} \times 9 \text{ slices}$, allowing for the observation of longer heterodyne beating characteristic without losing signal continuity.

The configuration parameters for each module are loaded, which contains data for operation, like the IP address, ports for connecting and calibrated values for the ADCs like the ramp duration of the Wilkinson ADC. A connection to the power supply of the Evalboards is established and restarted to get the TARGET modules in a known, clean hardware state after being idle or prior runs. A UDP connection to the modules is established. The modules are synced with synchronization pulses from the external trigger. The boards are initialized with pre-set settings for timing, operation, which voltage to use as pedestal and trigger settings. The voltage pedestal is a map of analog input to digital output values in ADC counts to ensure the desired output voltage despite manufacturing differences between each ADC. A warm-up time is waited for the ASICs to be on operational temperature. The external trigger provided by a frequency generator (TGF4082 by AIMTT) is switched on. Then data is taken with the trigger frequency of the external trigger. This is done via the UDP connection. The data is stored in the data path as an r0.tio file. The tio file is a special form of FITS (Flexible

Image Transport System [63]) file for TARGET ASICs. The data is calibrated with a dc transfer function and a pedestal. The dc transfer function measures the input and output voltage of the Evalboards under DC operation. This allows for the conversion of the ADC values to physical voltage values. The pedestal is a measurement taken without inputs, which is then subtracted from the measurement with inputs to account for background, biases and errors that stem from the TARGET ASICs architecture. Then the calibrated data is converted into several .npy files. One contains the data for all channels, one contains the TACK times, so the starting times of each event and one contains the FCIs, which are the first cell indices of each event. This is used for the calibration, since the ramp capacity of the Wilkinson ADC needs about 100 μ s to restore the default voltage value [64]. The samples in different blocks of the waveform are digitized with a different default value, which is why the FCI is needed to correctly identify in which block which cell took data.

6 Measurement introduction

In the last sections the theoretical background was detailed. The increase in sensitivity for third-generation gravitational wave observatories requires faster and more precise noise measurement systems. The fiber-array based phased camera offers a way to spatially resolve the lasers mode-content to analyze the misalignment of the laser in the cavity. The TARGET ASICs were introduced which are able to simultaneously sample and digitize 16 input channels for cost reduction. To analyze the wavefronts and mode-content of the laser a heterodyne measurement with a 80 MHz beating is taken and IQ demodulation is used to calculate the phases of each of the channels. The TARGET ASICs need to be tuned to increase the phase resolution for better mode decomposition. In this thesis multiple different methods to increase the phase resolutions were tested and analyzed. First the timing and synchronization were analyzed. Then the influence of different calibration methods on the phase resolution was evaluated. In the end, the sampling rate of the ASICs was reduced and the influence on the phase resolution examined. The data is located on the pc ecap-l045 and ecap-l066, and the analysis_scripts used are on ecap-l066 in the folder Software/et_scripts/Niklas_scripts.

7 Synchronization measurements

This chapter describes the influence of synchronization of the TARGET ASICs on the timing and phase measurement, which is necessary to accurately reconstruct the phase from the fiber-array based phase camera.

A strong requirement for the fiber-array phase camera is the simultaneous sampling of waveforms across multiple TARGET Evalboards at the same time without delay. To operate simultaneously, the boards must be synchronized to a common reference clock to eliminate relative timing drift between channels. To evaluate the effect the synchronization has on the measurement measurements with one Evalboard were done by injecting a 80 MHz sine wave from a GPS-synchronized arbitrary function generator (Aim TTi TGFA 4000) to the first input channel of the Evalboard. Then multiple events were taken with 500 ns and 512 ns trigger delay. The resulting sine waves were fitted. A histogram of the fitted phases is shown in Figure 25. Not synchronizing the function

generator to the Evalboard results in a large phase jitter, which indicates significant timing instability between the signal source and the sampling clock. Synchronizing the function generator to the clock of the Evalboard resulted in significantly narrowed the distribution which can be seen in Figure 25b. The standard deviation of the fit when synchronizing gets considerably smaller from 1.28 rad to 0.97 rad. This confirms that synchronizing the measurement devices with the TARGET Evalboards improves the temporal stability for wavefront reconstruction. Without this, the spatial phase map across the fiber-array would be deformed by relative timing fluctuations between the distributed TARGET modules.

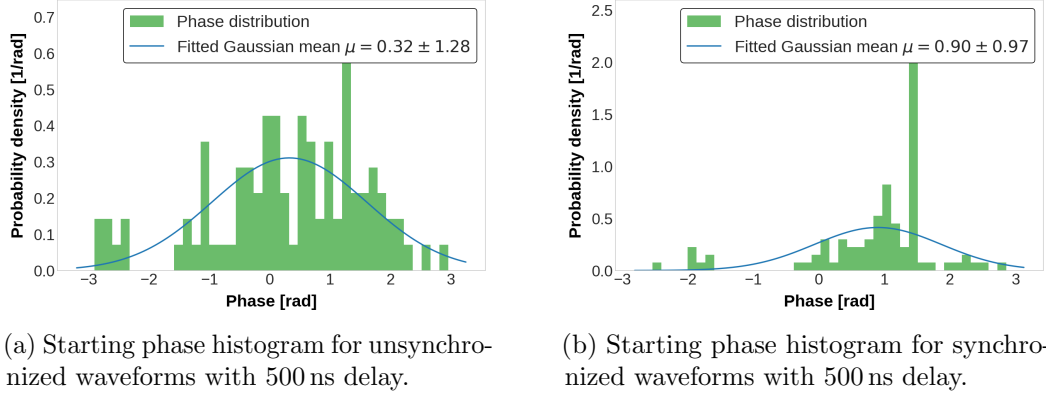


Figure 25: Histograms for synchronization measurements. The phase was retrieved by fits.

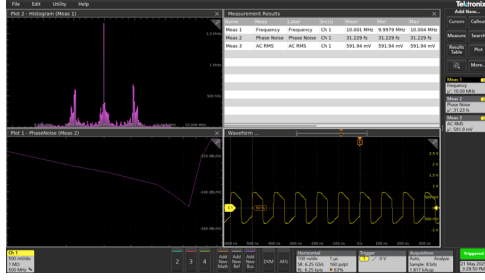
8 Clock jitter improvement

In this chapter different GPS-locked clock sources are analyzed to reduce the timing jitter of the TARGET ASICs.

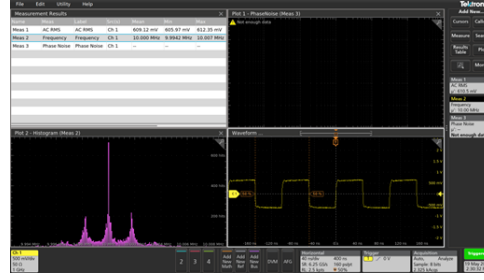
Accurate phase reconstruction in the fiber-array camera requires temporal stability of the system's clock. The Evalboards use an external 10 MHz reference clock, generated by a dedicated clock board, deriving the signal from a 125 MHz input source. To achieve this configuration, a pin originally designed as output pin was repurposed as dedicated clock input via a firmware modification.

To ensure optimal stability, two distinct GPS-locked clock sources were characterized. Using a digital oscilloscope, a comparative analysis of the clock's frequency stability, phase noise and RMS voltage fluctuations, with and without the clockboard was performed. The comparative result, summarized in Figure 27 demonstrate a significant performance difference between the two units. The clock sourced from the Electronics laboratory (ELab) showed higher stability, with a phase noise of 168.73 fs compared to 3.84 ps for the ET Lab reference clock and frequency an order of magnitude more stable.

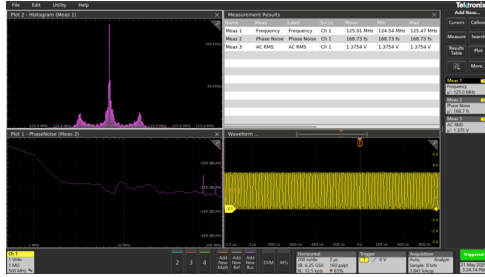
Figure 27 illustrates the spectral purity of the output clock. The ELab reference maintains a more precise frequency spectrum and lower noise floor in both cases. Based on these measurements, the ELab clock was permanently integrated as primary system clock into the ET Laboratory. This significantly improved the timing jitter, reducing the temporal variance from 0.09 ns to 0.06 ns. This timing jitter reduction is a critical



(a) Electronics Lab clock without clock board



(a) Einstein Telescope Lab clock without clock board



(b) Electronics Lab clock with clock board



(b) Einstein Telescope Lab clock with clock board

Figure 26: Electronics Lab clock measured with a digital Oscilloscope

Figure 27: Einstein Telescope Lab clock measured with a digital oscilloscope

improvement required for phase reconstruction to maintain the phase stability and coherence in the fiber-array phase camera, enabling the wavefront phase reconstruction required for accurate HOM content control in next-generation gravitational wave detectors.

		Electronics Lab Clock			ET Lab Clock		
		Mean	Min	Max	Mean	Min	Max
without clock board	Frequency	10.001 MHz	9.9979 MHz	10.004 MHz	10 MHz	9.9942 MHz	10.007 MHz
	Phase Noise	31.229 fs	31.229 fs	31.229 fs	not measured	not measured	not measured
	AC RMS	591.94 mV	591.94 mV	591.94 mV	609.12 mV	605.97 mV	612.35 mV
with clock board	Frequency	125.01 MHz	124.54 MHz	125.47 MHz	125.00 MHz	124.28 MHz	125.72 MHz
	Phase Noise	168.73 fs	168.73 fs	168.73 fs	3.8442 ps	2.4686 ps	7.2414 ps
	AC RMS	1.3754 mV	1.3754 mV	1.3754 mV	647.14 mV	646.13 mV	648.31 mV

Table 2: Clock Comparison: electronics lab vs ET lab (with/without clock board)

9 Calibration methods

This chapter discusses the influence of different calibration methods on the phase resolution for the fiber-array based phase camera.

To increase the accuracy of the digital output of the TARGET ASICs there are several methods to calibrate the data which are explained and their impact on the waveform

analyzed.

9.1 Voltage pedestal

Due to manufacturing variations, the SCA cells and the Wilkinson ADCs are not perfectly identical. The same applied voltage does not result in the same ADC output everywhere. A voltage pedestal is a hardware-level correction for the ADC/channel offsets. A lookup table maps DAC counts to VPED values measured with external ADCs in mV. This maps the desired output voltage to the specific DAC value required for each channel. 1500 mV is the preset voltage pedestal in the software to achieve consistency. The voltage pedestal for all 16 channels is shown in Figure 28. The VPed shows recurring jumps, when a large amount of bits flip. This can be seen at 2048 ADC counts as the binary transfers from 011111111111 to 100000000000. With more bits set to 1, more resistors are switched onto the circuit. The individual resistors tolerances stack up. When several resistors are switched off the network, this results in a voltage change seen as a jump [64]. This calibration is necessary for accurate phase demodulation, as the IQ demodulation depends on identical DC operating points across channels for accurate demodulation. It is not possible to access voltages above about 2.3V as the OTA (Operational Transconductance Amplifier) saturates [64].

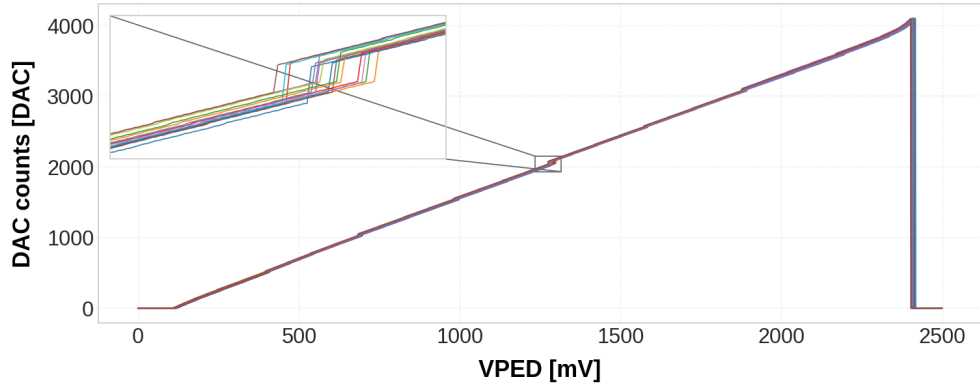


Figure 28: Voltage pedestal lookup of the Evalboard SN0002. Characteristic increase and

9.2 Pedestal calibration

A pedestal measurement quantifies the static noise floor of the acquisition chain under dark conditions, which means there is no input connected to the input channels of the Evalboard. Background contributions from electronics, ambient light, and thermal noise as well as the systematic errors introduced by the TARGET ASICs must be subtracted to isolate the optical signal. The pedestal is then subtracted from the measurement. The individual cells position in the different blocks of the waveform need to be account for in the pedestal. The Wilkinson ADC needs about 100 μ s to restore the default voltage value. For later blocks, the ramp starts at higher voltage values then in the first block, which leads to differences in ADC counts for the same cell voltage [64]. The characteristic peaks come from a timing mismatch of the TARGET ASICs, discussed in subsection 10.3. An example pedestal is shown in Figure 29

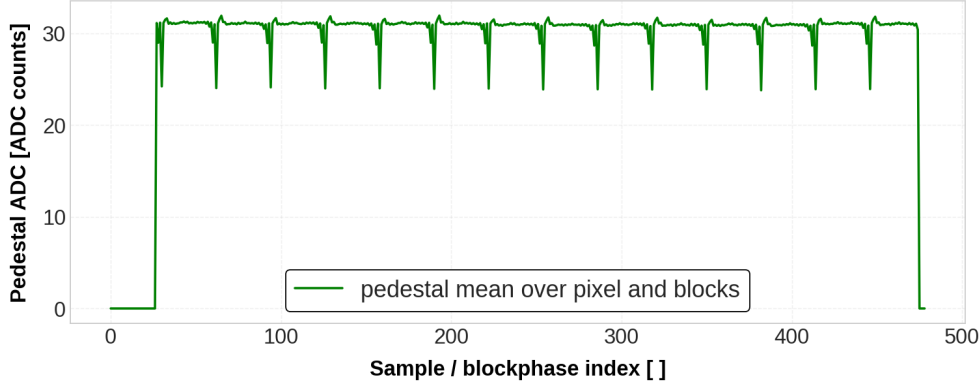


Figure 29: Pedestal for the Evalboard SN0002 with characteristic negative peaks at each 31 cell

9.3 DC - transfer function

The dc transfer function accounts for the influence of different dc voltage levels on the output of the TARGET Evalboards. This script loads the vped lookup table and generates a list of voltage values in 50 mV increments. It connects to the TARGET modules and for each voltage value it looks up and programs the DAC values for each channel and ASIC. Then it enables the hardware trigger and takes 5 s of data for each voltage value. It saves each measurement in a different file and writes the filenames in a text-file. This file can then be used to generate the dc-transfer function. An example dc-transfer function is shown in Figure 30. One can see the dc-transfer function for every pixel and cell in the TARGET ASIC.

The influence of the dc-transfer function on the phase resolution was analyzed by calibrating the same dataset with and without the dc-transfer function.

The data needs to be loaded with numpy and the IQ demodulation is done. The analysis starts with loading the mapping of the pixels of the fiber array to the channels of TARGET. The TACK timestamps are normalized relative to the first timestamp, corresponding to the first trigger and then a continuous time array in ns for each event is build. The waveforms are then cut into segments corresponding to the individual events from one multi-triggered event. In Figure 31 one event from a dataset taken with 1 GSa/s sampling rate and multi-trigger 8 is shown in blue with the corresponding cut data to multievent 7 with the constructed time base of 1 ns/Sa (orange). This shows the Evalboard works to take data at 1 GSa/s sampling rate, and the cutting of the data and the time base generation also work like expected.

The active pixels from the mapping are filtered and for each pixel and each sideband the IQ demodulation, described in subsection 4.3 is performed. For each event from the cut waveforms the in-phase I and quadrature Q values are computed. From the IQ values the phase and amplitude are calculated as explained in subsection 4.3. The time for each events phase and amplitude point corresponds to the TACK time. The output for each pixel is stored with the mapping, sidebands, time array in ns, the phase arrays of each sideband and the amplitude arrays for each sideband. After the main analysis is done, the phase is unwrapped to be continuously in a 2π range. Then the phase resolution is calculated as the standard deviation of the phase, the signal power is calculated from the mean of the amplitude and then converted to dBm. The unit dBm

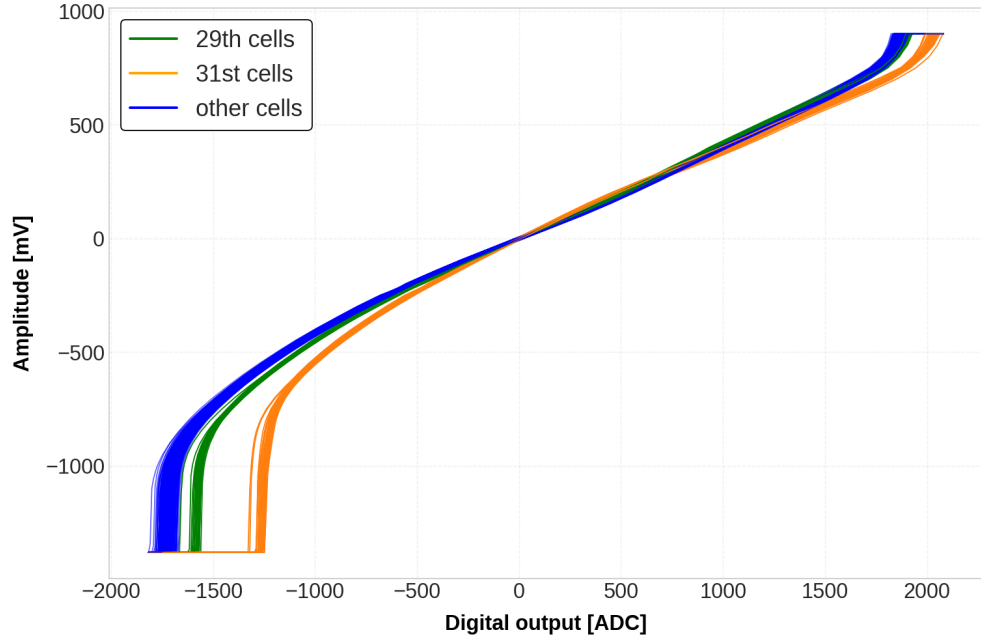


Figure 30: 4096 individual dc transfer functions of channel 0 of one block phase for SN0002, the individual behaviors are color-coded based on the cells position in the block

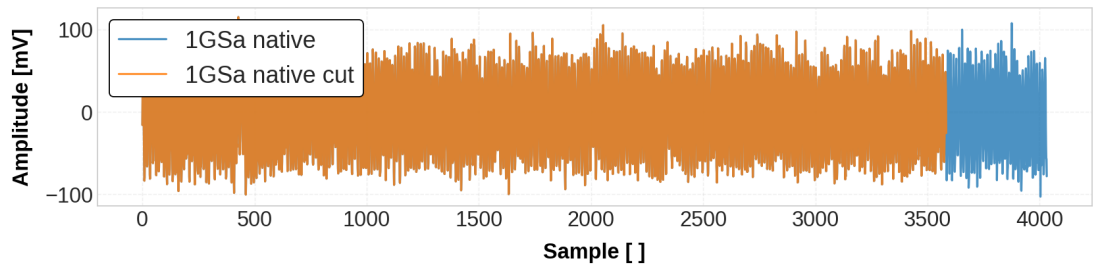


Figure 31: Calibrated data (blue) and cut data (orange) from one event taken with 1 GSa/s sampling rate

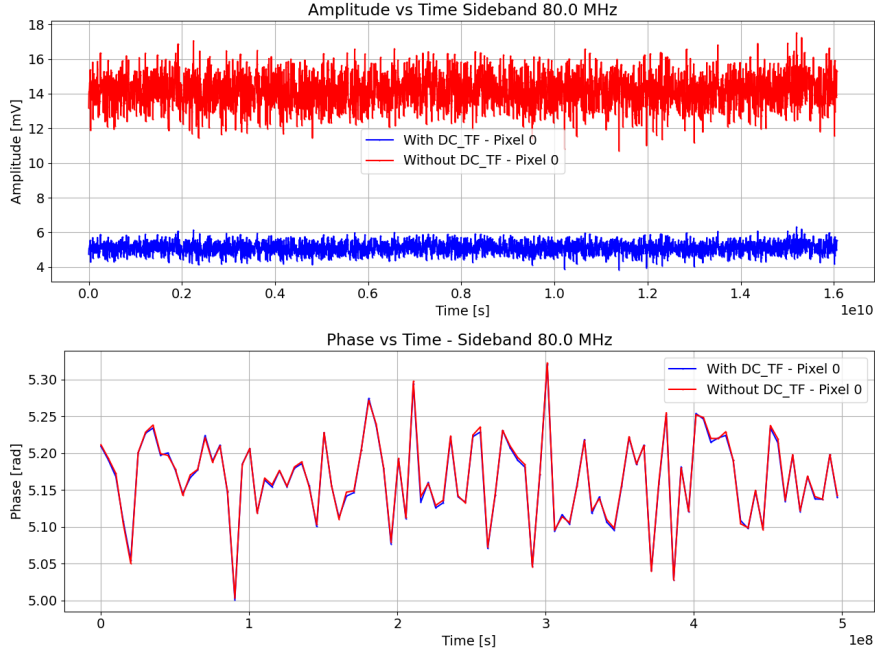


Figure 32: Difference of amplitude and phase with and without dc transfer function for pixel 0

is a logarithmic unit of measurement of the absolute power based on 1 mW. Unlike the unit dB, which just describes a relative power ratios, dBm specifies a concrete power level. The power in dBm can be converted from mW with the following formula [65]:

$$P(dBm) = 10 \cdot \log_{10}\left(\frac{P(mW)}{1mW}\right) \quad (19)$$

These were plotted with and without dc-transfer function. An example amplitude and phase plot for pixel 0 and sideband 80 MHz can be seen in Figure 32. For each pixel and sideband the resolution was calculated as the standard deviation of the phase. This resulted in the plot, which can be seen in [Figure 33]. The red and blue points are the resolutions without and with dc-transfer function respectively. The dotted black line is the goal which the phase resolution should reach. The purple line is marks the integration time dependent part of the resolution. One can clearly see that the phase resolution is improved with the dc-transfer function, showing the need for calibrating the data with the dc-transfer function to accurately determine the phase with maximal resolution.

10 500 MSample/s sampling rate

In this chapter the implementation of the reduced sampling rate and the analysis of the sampling rate on the phase resolution is discussed.

It was proposed to reduce the sampling rate from 1 GSa/s to 500 MSa/s, to reduce data load and increase the integration time for the IQ-demodulation. This increase in integration time is coupled to an increase in phase resolution due to bandwidth reduction and increased averaging. This is analogous to other data acquisition systems,

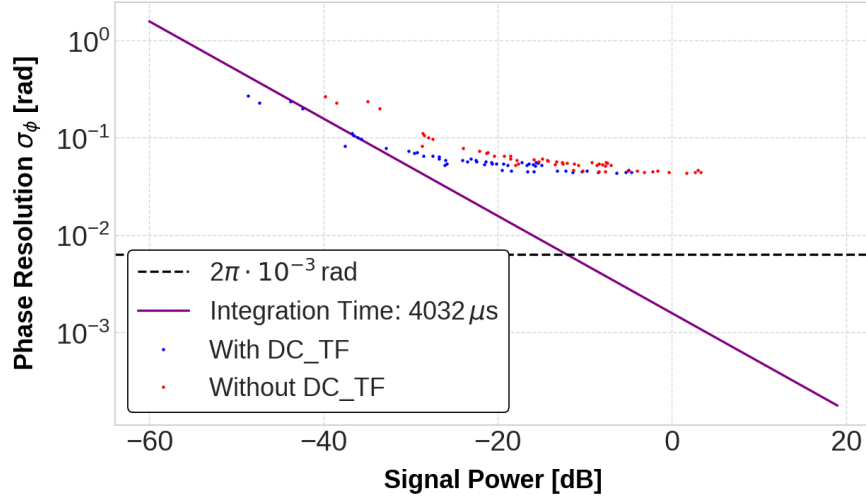


Figure 33: DC TF Influence on resolution

where an increase in measured time for one data point results in a higher signal to noise (SNR) ratio such as MRIs [66].

To estimate the effect of the reduced sampling rate on the phase, phase resolution and amplitude a dataset taken with the laser setup at 1 GSa/s was evaluated. The data was written into two arrays. In one every second sample was cut to create a dataset with an artificial 500 MSa/s sampling rate. Then each dataset was cut into the segments corresponding to the events of 4032 samples and for each segment the corresponding time was generated for the 1 GSa/s with 1 ns spacing for the 500 MSa/s with 2 ns spacing. One time the 500 MSa/s data array was cut to 4032 samples per event, one time to 2016 samples to simulate the effect when not changing the integration time. Then IQ-demodulation as described in subsection 4.3 was performed for every segment to create one phase and amplitude point per event. The phase resolution was calculated as the standard deviation of the phase for every pixel and sideband. The resulting phase resolution was plotted over the signal power from the sidebands, which can be seen in Figure 34. The signal power is calculated from the mean of the amplitude for each channel and sideband and then converted to dBm. This shows that a reduction of sampling rate lowers the phase resolution, proving the concept and showing that a reduction of sampling rate is needed. Although lower sampling rates than 500 MSa/s didn't show much improvement as shown in Figure 34, where the 250 MSa/s didn't bring significant improvement compared to the 500 MSa/s.

10.1 Rclod

One idea to achieve the 500 MSa/s sampling rate was by adding an additional capacitor in the ASIC-internal time cell capacitor switch during the measurement. This increases the capacity and therefore the delay for each time point. This was achieved with the command `WriteTARGETRegister(True,False,False,False,False,True,0x4b,4)`, which writes the value to enable the capacitor directly in the TARGET register.

Unfortunately that didn't produce the aspired 500 MSa/s. An example dataset that resulted from the rclod can be seen in Figure 35. One can see some parts that look like

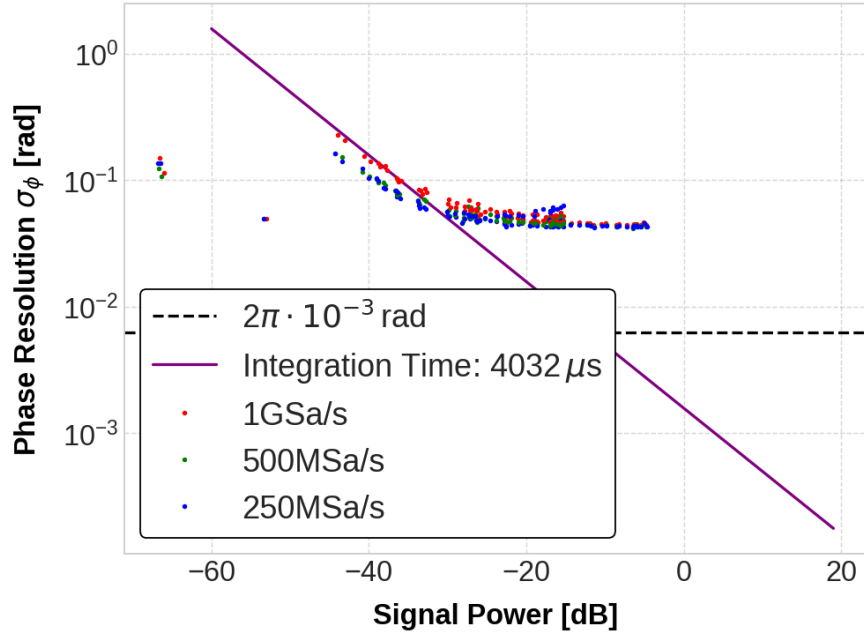


Figure 34: Phase resolution over signal power with dataset taken at 1 GSa/s and theoretical 500 MSa/s and 250 MSa/s

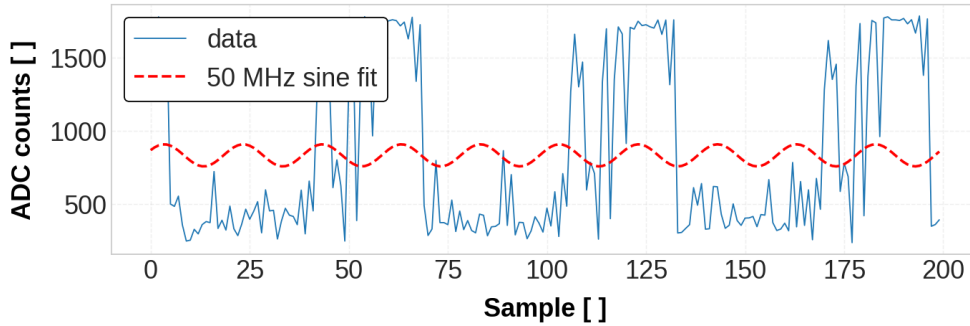


Figure 35: Data with Rload on and a wrong timing configuration

the waveform, but they are cut off by sections that are zero. This behavior could be fixed by changing different timing parameters. There are several control supplies and control delays that needed to be tuned. The control delay on low to high transition of the sampling delay circuit, called $vadjN$ was changed. This resulted in clearer waveforms, which can be seen in Figure 36. Although the waveform wasn't cut off now it still showed that the sampling rate wasn't 500 MSa/s, since the period of the 50 MHz sine wave corresponded to 20 samples, which corresponds to 1 ns per sample with a sine wave period of 20 ns. Therefore this method of enabling the additional capacitor didn't produce the desired 500 MSa/s.

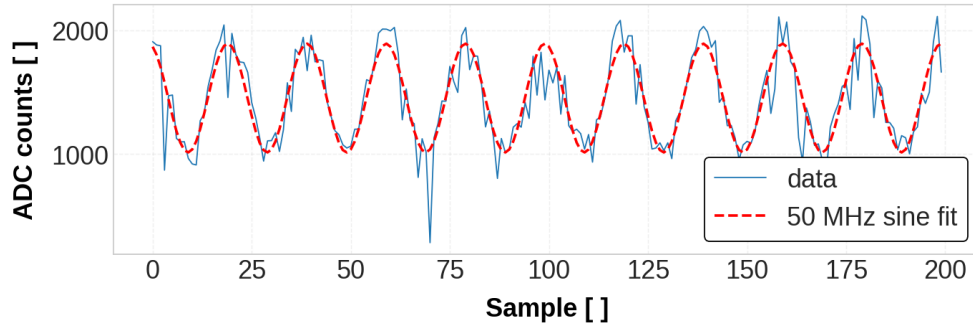


Figure 36: Rload data with changed vadjN timing parameter

10.2 sstclk

Another idea to achieve a sampling rate of 500 MSa/s was to let the sstclk that goes out to the TARGET chip, run at half speed. The sstclk is a clock driven by a sampling counter. The SST_MAIN_pi4 uses that as input for a phase-locked-loop to derive internal clocks, like the digitizing clock or the Wilkinson clock. At each rising edge of the sst clock, a different signal called sstin_p for positive sst clock is advanced which is send to TARGET. This signal advances the internal write pointer, that connects the analog input to the next sampling cell and writes this cell. Changing the sstin_p only at the rising edges only advances the signal every second step, but since the sst clock still advances every step, this causes alternate cells to be skipped. This was achieved by implementing a flip-flop into the top-level vhdl code. This flip-flop takes the sst clock as input and toggles on each rising edge of this clock. This creates a clock at half the speed as the input clock. The vhdl code is shown in Codeblock 1.

```

1 sstin_invert <= not(sstin_i) and not(add_control(15) and ramp_i); --
  lvds in is inverted, but maybe also on TC
2 Process(sstin_invert)
3 Begin
4   if (rising_edge(sstin_invert)) then
5     sstin_slow <= not(sstin_slow);
6   end if;
7 End Process;
```

Codeblock 1: sstclk flip-flop

The additional Rload capacitor was switched on to achieve the correct time delay. After that the timing parameters in the software needed to be changed. The WR_ADDR.Incr*_Delay controls when the memory write address advances, which is responsible for sample spacing. The SSPIN controls the timing of the sample select pulse timing, controlling which samples to process. The SSTIN controls the sample strobe clock and therefore manages when to capture data. The WR_STRB*_Delay controls when data is written to alternating buffer storages. The relative timing between the parameters for optimal function can be seen in Figure 37. This approximate timing needs to be maintained to ensure accurate data taking. The correct timing parameters resulted in a working 500 MSa/s sampling rate. A new dc transfer function with the firmware for 500 MSa/s was taken, which can be seen in Figure 38. One can see that many cells don't have the typical shape of the dc transfer function like in Figure 30 for 1 GSa/s. There is no data written into these cells, because of the way the firmware is

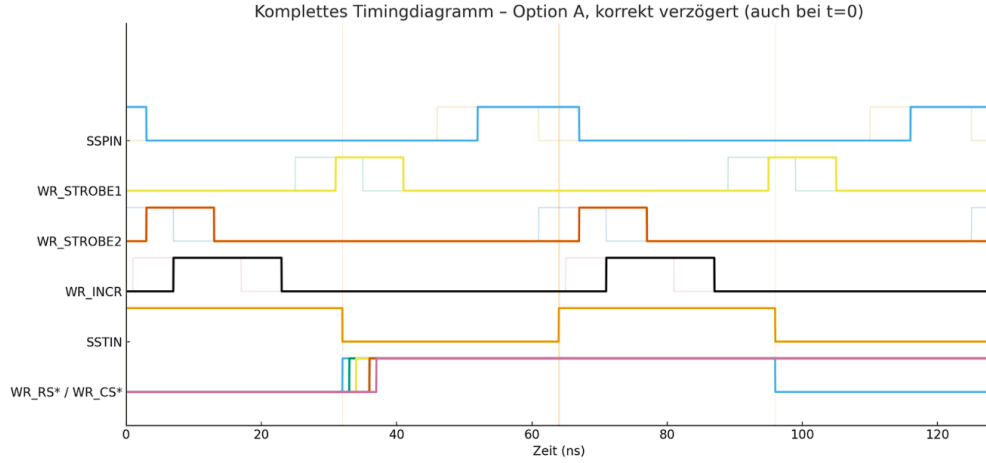


Figure 37: Timing parameter diagram [67]

written. Only every second cell is skipped by the address pointer, this creates empty cells. There the DC transfer function reduces to the pedestal of that cell. Only near the extreme cases around ± 1000 the TARGET cells leaves that plateau due to saturation or leakage.

10.3 Addressing error measurement

With the 500 MSa/s sampling rate there was an error in each waveform. To determine this, data was taken with different kind of waveforms. First a 10 MHz sine wave was taken to determine samples that appear twice, which would point to an addressing error. The wave can be seen in Figure 39. It can't be determined from that if there is an addressing error, so different datasets with ramp waveforms of different frequencies were taken. An example with a frequency of 2.5 MHz is shown in Figure 40. Here the SSPINLE delay is a timing parameter which controls the leading edge of the. With a higher SSPIN delay of 49 a ramp with the same frequency the spike error can't be seen anymore. So the error was a timing error, as the sspin controls which sample to read. If this timing is delayed a wrong sample is read. This proved that the correct timing could resolve the error, which is necessary for accurate phase reconstruction.

10.4 Time base measurement with 10MHz

To further analyze the time base stability for the 500 MSa/s sample rate a 10 MHz sine wave generated by a function generator that was not synchronized to the TARGET Evalboards was measured. The measured waveform was then analyzed to generate a time base, in this case calculated as relative width of each sampling cell in ns. To do this, first the mean was subtracted. Then rising-threshold zero-crossing points were found and mapped to global storage-cell IDs using the FCI information. This works by dividing the FCI by the number of storage cells (64) as modulo. A 64-bin histogram was built of how often each cell was hit at those zero-crossings. Then the histogram is reinterpreted as relative widths. This only works with a sine wave generated by a function generator not synchronized to the TARGET ASIC clock. This creates a sine wave with a phase that is effectively random over many events, which creates uniformly

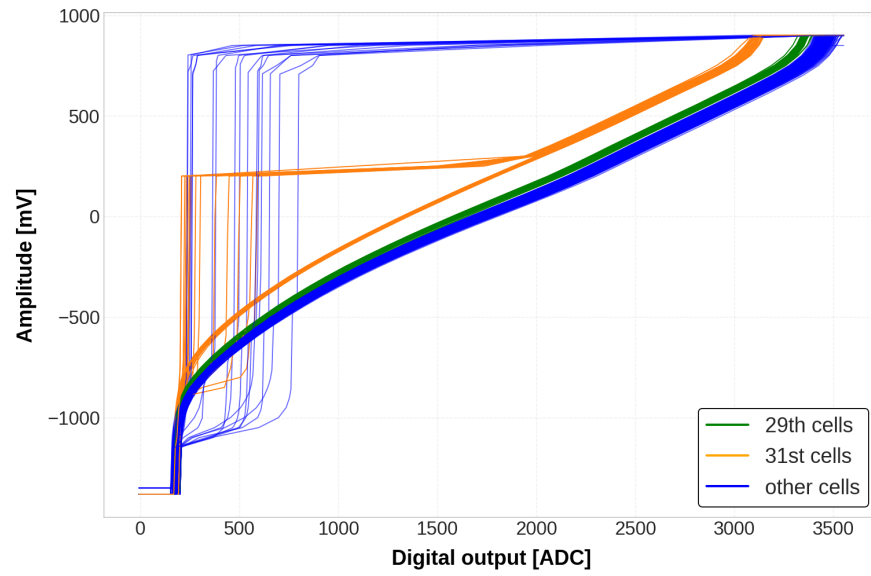


Figure 38: DC transfer function for 500 MSa/s sampling rate

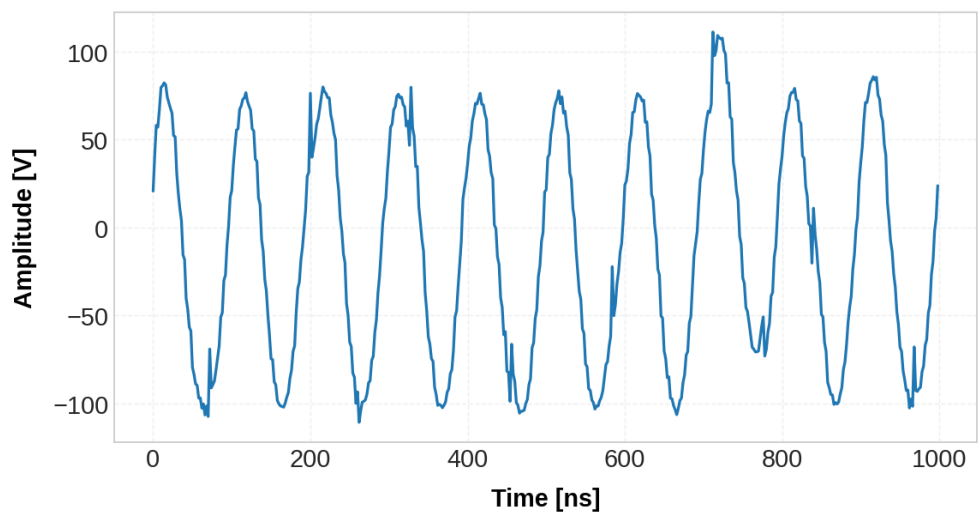


Figure 39: Sine wave of 10 MHz to determine the nature of the spike error

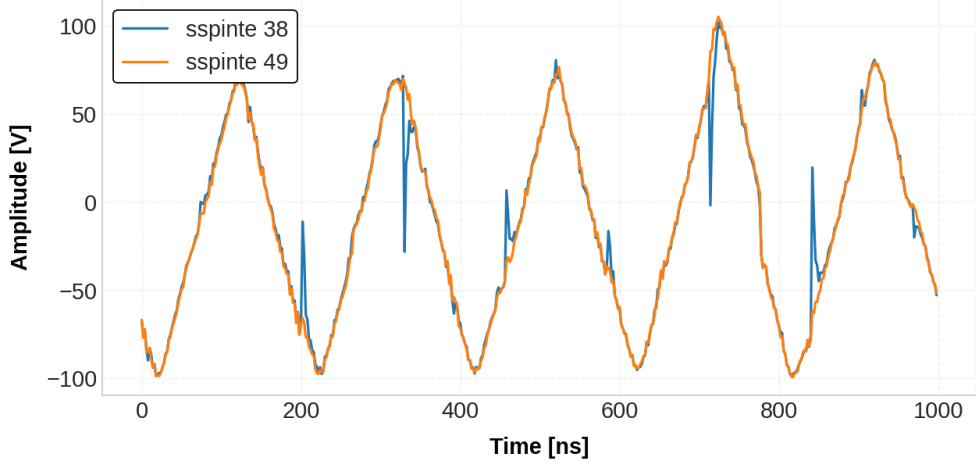


Figure 40: Comparison of ramps with frequency 2.5 MHz and sspin delay of 38 and 49

distributed zero-crossing times over the sampling time. A sampling cell with a larger true time width occupies more time, so random crossings land in it more often. The time per cell i dt_i therefore is proportional to the zero-crossing counts for that cell N_i . The counts are converted to widths by normalization with the sum of all counts $\sum_j N_j$ and the total time of one sampling window T_{total} which is 128 ns for 500 MSa/s. The formula is then given by Equation 20.

$$dt_i \approx \frac{N_i}{\sum_j N_j} \cdot T_{total} \quad (20)$$

This resulted in the relative widths for the each sampling cell which can be seen in Figure 41. The time of one sampling window and the walk-off, the difference to the uniform time base, shown in Figure 42, show that the time difference at the end of the sampling window goes beyond 3 ns and then decreases to 2 ns. We therefore need to optimize the timing parameters further.

10.5 Timing optimization

To further optimize the timing, measurements with a 50 MHz sine wave were taken to optimize the timing of the TARGET ASICs. Fits on the 64 sample wide windows, from the SCA, showed a deviation from the 50 MHz, which means that the TARGET ASIC's sampling rate was not perfectly 500 MSa/s. The frequencies from the fits were not 50 MHz but higher, ranging from 52 MHz at an sstfb of 59 to above 54 MHz at lower sstfb values. This shows the sampling rate was lower than 500 MSa/s, ranging from 460 MSa/s for lower sst_outfb values to 480 MSa/s for higher sst_outfb values. A frequency histogram from fits on measurements of a sine wave with 50 MHz frequency is shown in Figure 43, which showed the frequency shift.

The timing parameter sstfb_outfb needed to be optimized. The sstclock is an external reference clock from the FPGA. The sst_outfb is a delayed version of the sstclk which is then used in a delay locked loop style timing loop also involving the parameter vadjN, tuned in subsection 10.1 and is used to generate the sample strobe timing and to stay in phase with the sst clock period. The additional capacitor Rload, added to achieve

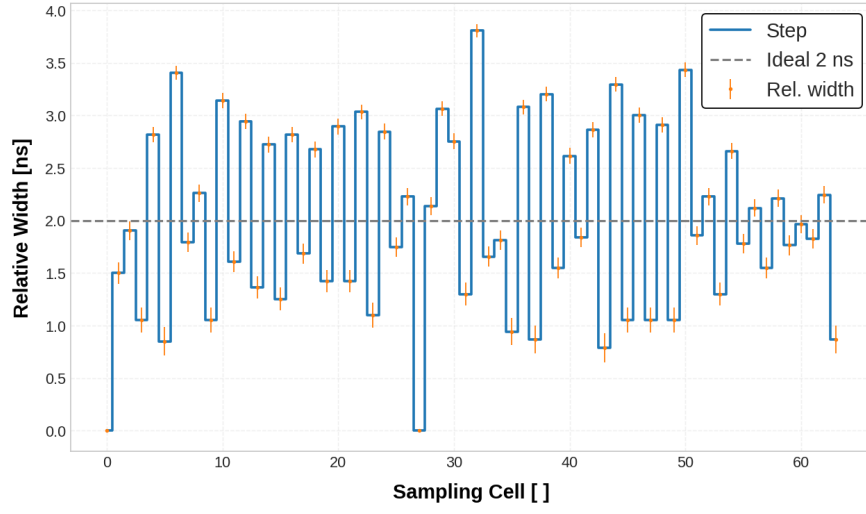


Figure 41: Time base calculated by normalization of histogram of zero-crossings across sampling cells

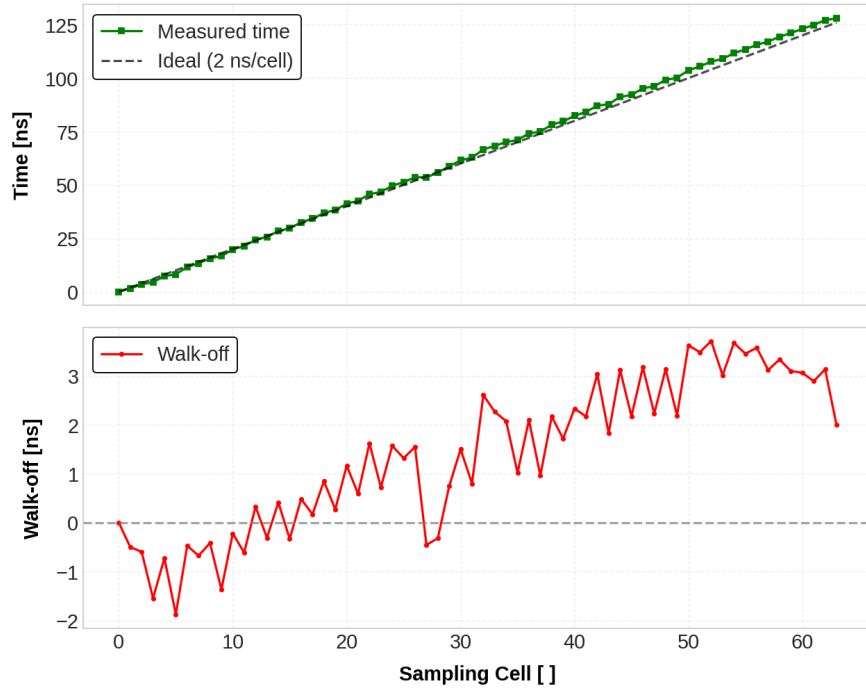


Figure 42: Time base cumulative time over walk off 10MHz. The walk off is the difference between a uniform 2 ns steps time base and the time base measured.

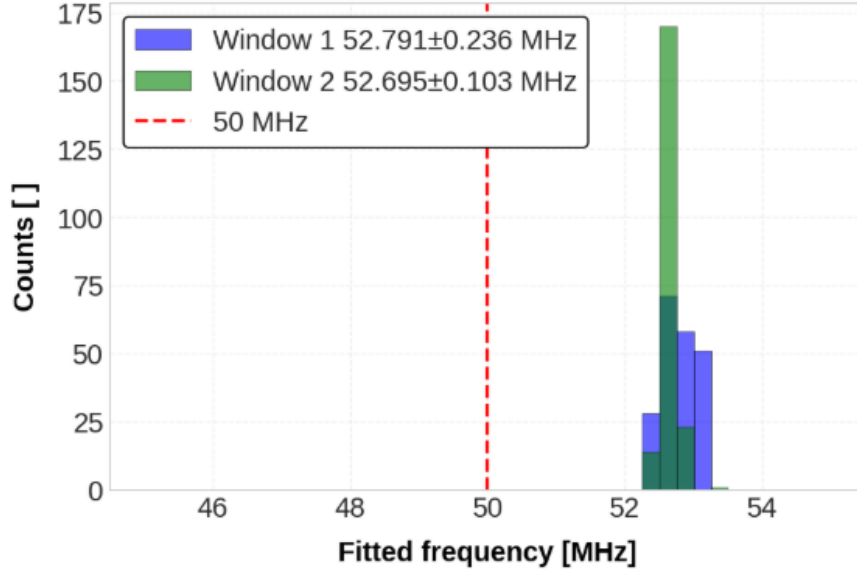


Figure 43: Frequency histogram from fits showed a higher frequency than the input waves frequency

the 500 MSa/s sampling rate, also adds an internal delay in the timing. Due to this reason the `sst_outfb` parameter needs to be lower than for the 1 GSa/s version, which was set to an `sst_outfb` value of 63. The waveforms for different `sst_outfb` values is shown in Figure 44. For `sst_outfb` values lower than 54, the timing drifts from the `sstclk`, and the samples are taken from random places, which produces a noisy waveform. For `sst_outfb` values between 55 and 60, the timing doesn't drift significantly from the `sstclk` and the waveforms are clearly visible. Although unlike with the 1 GSa/s sampling rate, the errors, resulting in periodically downward peaks, were not reduced by the calibration, but are visible as periodic peaks in the waveform. This error is discussed in subsection 10.3. For higher `sst_outfb` values than 61, the timing drifts again, resulting in a noisy waveform. This is shown in Figure 44.

A sine wave with a frequency of 50 MHz was measured with different settings of `sstfb` and `vtrimT` to determine the correct set of parameters to achieve the right timing. For 1 GSa/s the values were set to 56 and 1107 for the `sstfb` out delay and `vtrimT` parameter.

For 500 MSa/s a range of 55 - 64 for the `sstfb` out delay was tested with `vtrimT` parameter values ranging from 800 to 4000. An `sstfb` out delay value greater than 61 resulted in a waveform consisting of just noise. The datasets were calibrated with a uniform 2 ns based time base and a new time base to analyze the influence of the time base on the timing stability. This new time base was generated for every dataset through the mechanism described in subsubsection 10.5.1. The time base correction proved to be working by measuring the frequency histogram. In Figure 45 a histogram of the fitted frequencies for `sst_outfb` is shown. On the left with the uniform 2 ns steps time base and on the right with a corrected time base calculated through the mechanism described in subsubsection 10.5.1. The time base correction shifted the frequency of the sine wave to 50 MHz, which proves the time base correction is necessary to accurately determine the frequency and phase of the laser.

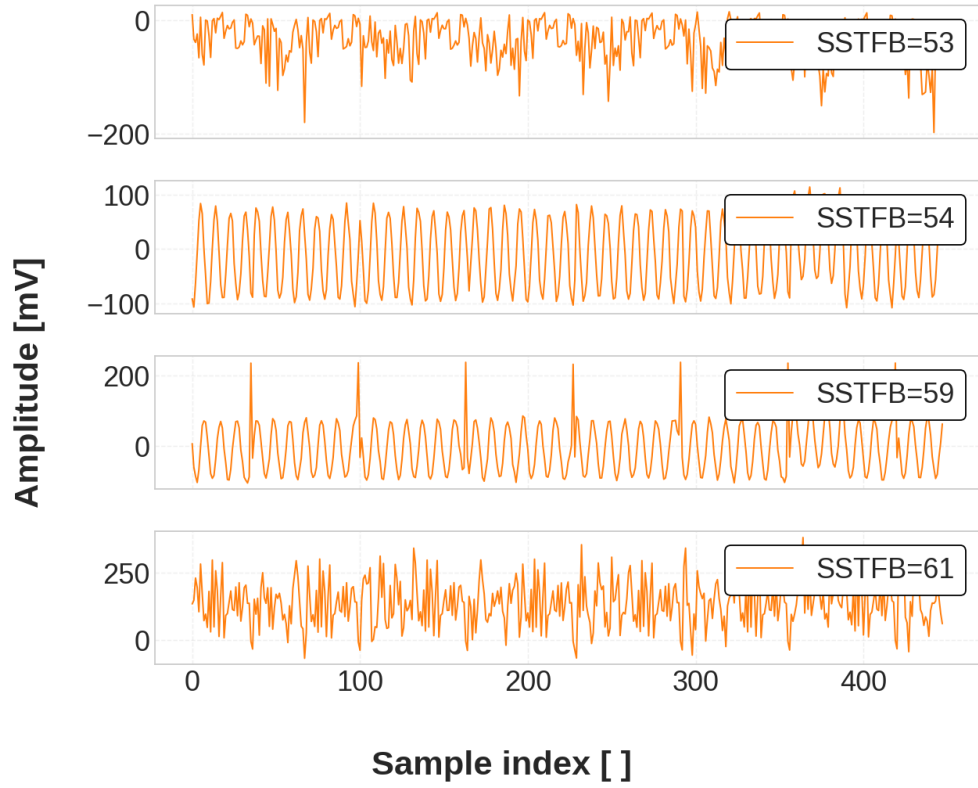


Figure 44: Waveforms taken with a 50 MHz input with different sst_outfb values.

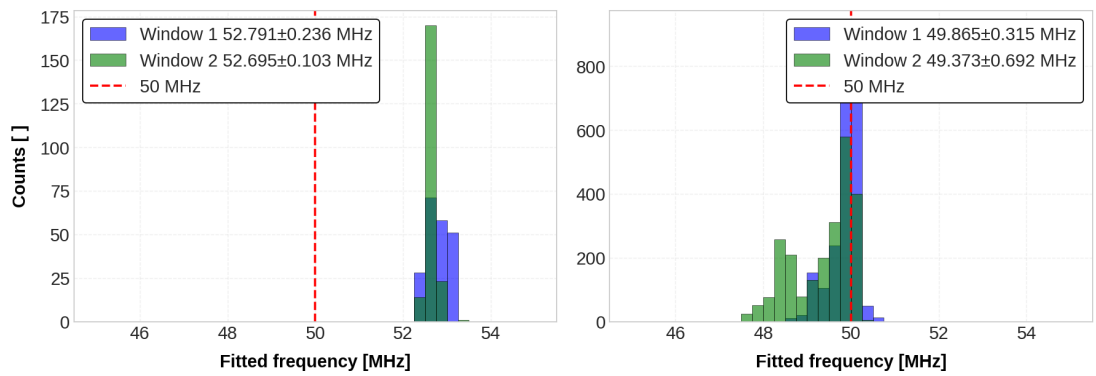


Figure 45: Frequency histogram for the uniform time base (left) and the corrected time base (right), showing the time base correction resulted in correct frequencies.

When the two parameters `sstfb` and `vtrimT` are optimal the phase difference of consecutive 64 sample windows is reduced to zero. The phase difference was calculated with a sine fit to each window and then the resulting phases were subtracted. The result from the different measurements with different `vtrimT` and `sstfb` are shown in Figure 46. When `sstfb` is 56, there is a zero crossing with the uniform time base. With the non uniform time base there is no zero crossing. This is contrary to the frequency shift. The time base correction actually decreases the accuracy of the timing. This shows the time base fluctuations are just in the 64 sampling wide windows, and inter windows the time-base correction actually worsens the shift.

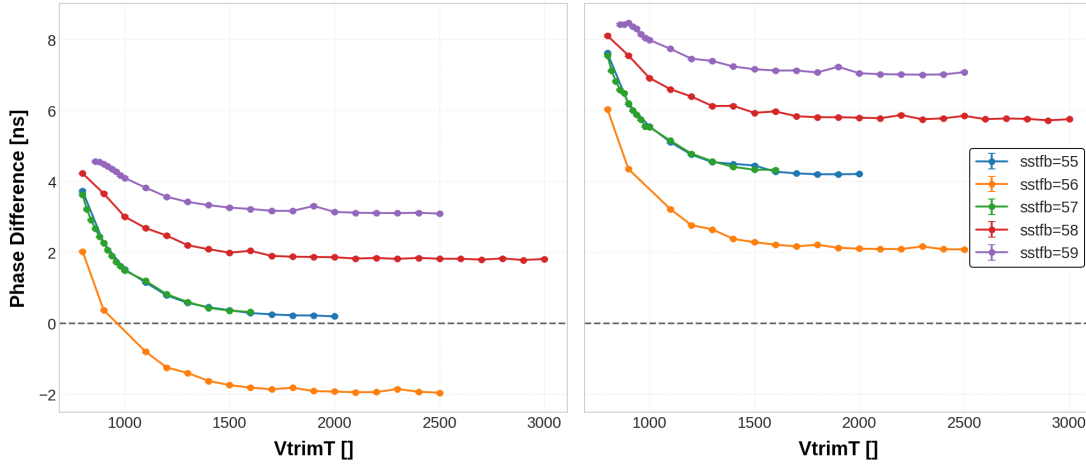


Figure 46: Phase difference over `vtrimT` with different `sstfb` parameter values

10.5.1 Time base correction

The measured data was fitted with a sine wave of the form

$$\text{ sine}(x, y, A, \theta, c) = A \cdot \text{ sine}(2 \cdot \pi \cdot \omega \cdot x + \phi) + C \quad (21)$$

Then for each sample in a 64 sample window the nominal time difference of the time base of 2 ns/Sa was shifted to correctly overlap the a sine wave of 50 MHz with the same phase and amplitude. The 64 sample wide windows come from the architecture of the SCA, described in subsection 5.2.

A function was written to estimate the time difference to shift each sample to correctly represent a sine wave of 50 MHz. It constructs a corrected time by adding a small shift dt_{ns} to the nominal time $t_{uncorrected}$ Equation 22.

$$t_{corrected} = t_{uncorrected} + dt_{ns} \quad (22)$$

Then the function computes the amplitude and phase of a model 50 MHz sine with the same amplitude and phase of the fit. Two phases are computed from the measured waveform at the time $t_{uncorrected}$, one with the arcsine function, and the other is phase shifted by π . It then chooses the phase that is closer to the 50 MHz sine waves phase and computes the amplitude and phase difference. With the `minimize_scalar` function from `scipy` the time difference dt_{ns} is then changed to minimize the amplitude and phase difference of the measured waveform and the sine wave of 50 MHz. This time difference is then applied to the corresponding sample index time in the time base.

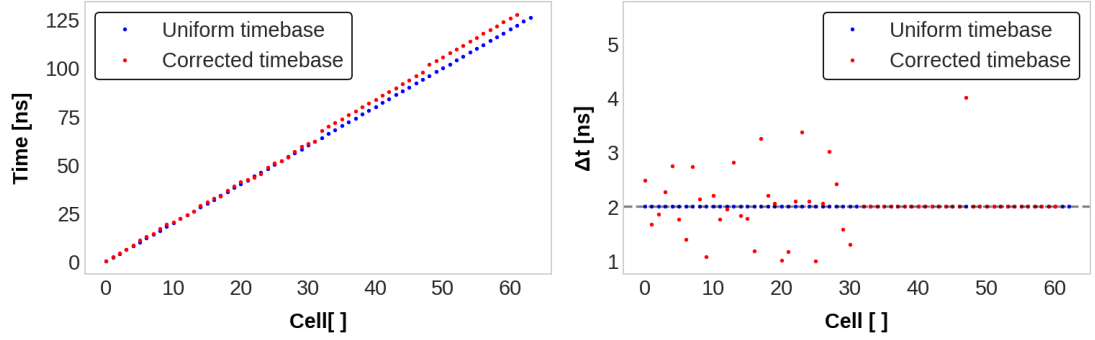


Figure 47: Time base differences between the uniform and corrected , Left: Cumulative time in one 64Sa window, Right: Time difference between consecutive time points

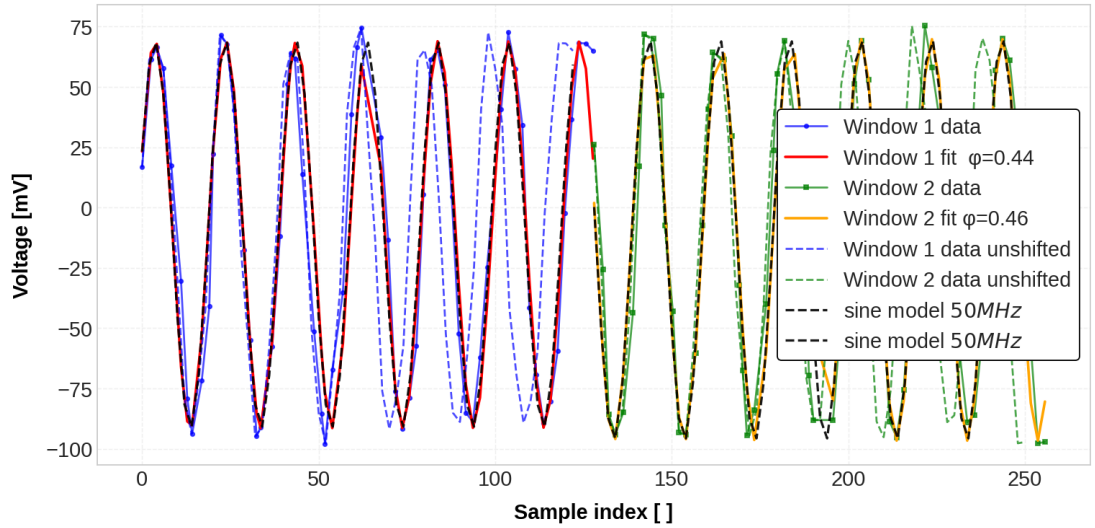


Figure 48: Waveform with uniform time base in dotted lines for each window and with corrected time base in solid line, in black dotted the sine with a frequency of 50 MHz is shown.

How to apply the time base to measurement data When analyzing data the time base needs to be applied to each 64 samples wide sampling window. This is done with a version of the already existing waveform cutter. The cutter was previously used to cut data into the integration windows and generating a uniform time base for each. It was modified to take in a custom time base and apply it to each window separated by 128 ns. It also cuts data points taken in a 64 sample window that are beyond 128 ns and the errors mentioned in subsection 10.3. The time base correction on the 50 MHz sine wave shifted the frequency histogram from fits to be centered around 50 MHz shown in Figure 45.

10.6 500 MSa/s sampling rate laser measurement

Measurements with the laser setup were taken to analyze the true influence of the 500 MSa/s sampling rate compared to the theoretical 500 MSa/s sampling rate derived

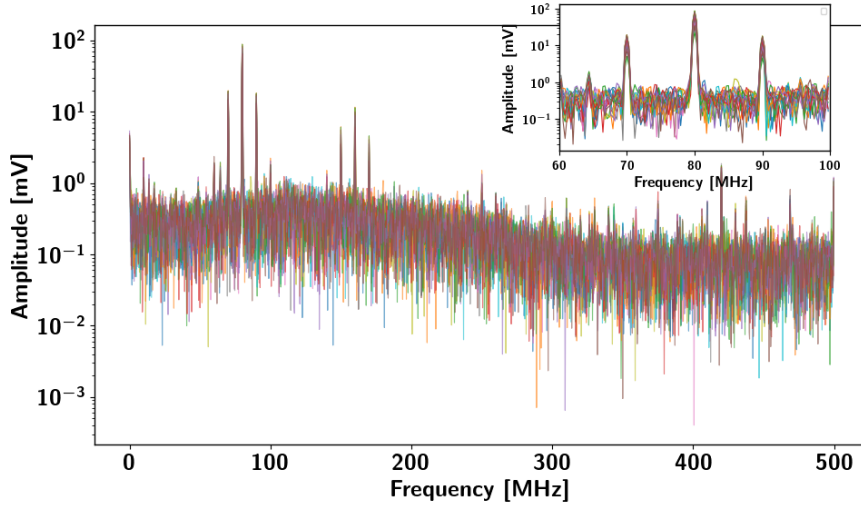


Figure 49: Power spectral density of the data taken with 1 GSa/s sampling rate

from a measurement with 1 GSa/s sampling rate. One dataset was taken with the latest working firmware with 1 GSa/s sampling rate and one dataset with the firmware for 500 MSa/s sampling rate. For both waveforms a power spectral density was calculated by Fourier transform. The power spectral density can be seen in Figure 49 and Figure 50 for 1 GSa/s and 500 MSa/s respectively. The power spectrum for 1 GSa/s sampling rate shows frequencies up until 500 MHz while the spectrum for 500 MSa/s shows frequencies up to 250 MHz. This is due to the Nyquist theorem, which states that the highest frequency that can be resolved with a given sampling rate is half that sampling rate, which can be expressed with the formula Equation 23 [68]. The carrier frequency is at 80 MHz and the sidebands are modulated at 10 MHz around the carrier frequency and therefore it is possible for both sampling rates to fully resolve all the relevant frequencies. In the 500 MSa/s there are many peaks visible at different frequencies. This is a result of the timing error that is still measured, but they are not of interest at the moment, since we only look at the specific frequencies of the carrier and sidebands. Looking at just the window from 60 MHz to 100 MHz visualized in the corners one can clearly see the peaks from the carrier and sideband frequency in the psd from 1 GSa/s sampling rate, but in the power spectrum of the data taken with 500 MSa/s, the peak at 70 MHz is too broad to stem only from the sideband modulated onto the carrier frequency. So there must be a component of the addressing error in subsection 10.3.

$$f_s = 2 \cdot f_{max} \quad (23)$$

The influence of the time base correction was analyzed by demodulating at frequencies near the carrier and sideband frequency to visualize a frequency and amplitude shift. The frequency steps at which the data was demodulated was chosen based on the sampling frequency divided by the number of samples. This resulted in frequency steps of 124 kHz, starting at 0 MHz.

$$\Delta f = \frac{f_{sampling}}{N_{Samples}} = \frac{500 \cdot 10^6 \text{ kHz}}{4032} = 124 \text{ kHz} \quad (24)$$

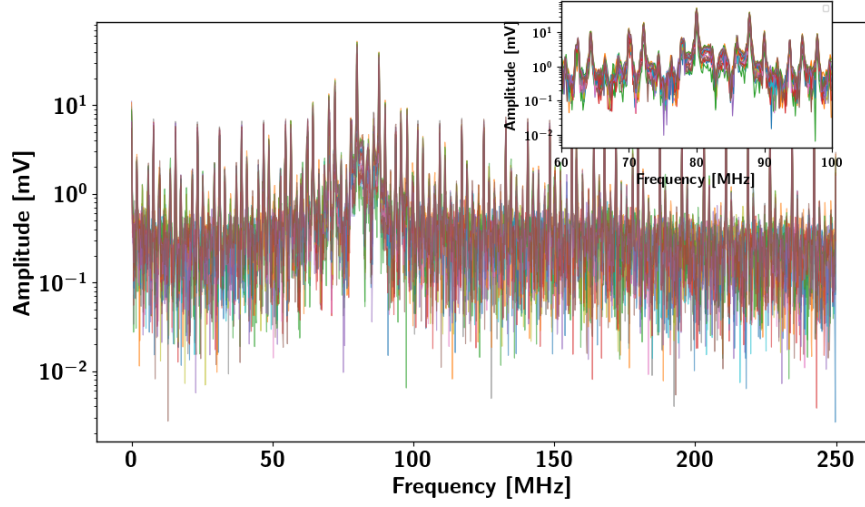


Figure 50: Power spectral density of the data taken with 500 MSa/s sampling rate

In Figure 51, Figure 52 the resulting power for each frequency band is shown. In blue with a uniform time base of 2 ns steps, and in red the corrected time base. One can see that the only frequencies where the power is amplified by the new time base are the carrier and sidebands frequencies. The surrounding frequencies are decreased in power, which mostly comes from the amplitude cutting to cut the addressing error. This proves the time base is correctly implemented to reduce the errors of the waveform, but there is no frequency shift observed, which would be expected.

For visualization for the pixel with the best and worst phase resolution the phase and amplitude are shown in Figure 53 and Figure 54 respectively. One can see that the phase and amplitude in Figure 53 are much less noisy than the ones in Figure 54. The first image shows the phase and amplitude for the main carrier frequency at a pixel with high amplitude, while the other image shows a pixel with low amplitude at the sideband frequency 90 MHz. The sideband 90 MHz also fell right into a frequency range, where a peak is located which stems from the addressing error. One can also see that there was a drift in the phase and amplitudes, which is plotted as blue points. This could be a thermal drift. Therefore calculating the resolutions based on the original phase and amplitude yielded worse phase resolutions. Consequently phase and amplitude time evolutions were fitted with a polyfit. A third order polyfit was chosen for the phase and a second order one for the amplitude, since the third order fit for phase errors showed lower residuals in the phase error compensation method from [69] which also used IQ demodulation, but second order were already enough. The fit is visualized in red. Then the deviation of the fit from the mean was subtracted from corresponding phase and amplitude time evolutions.

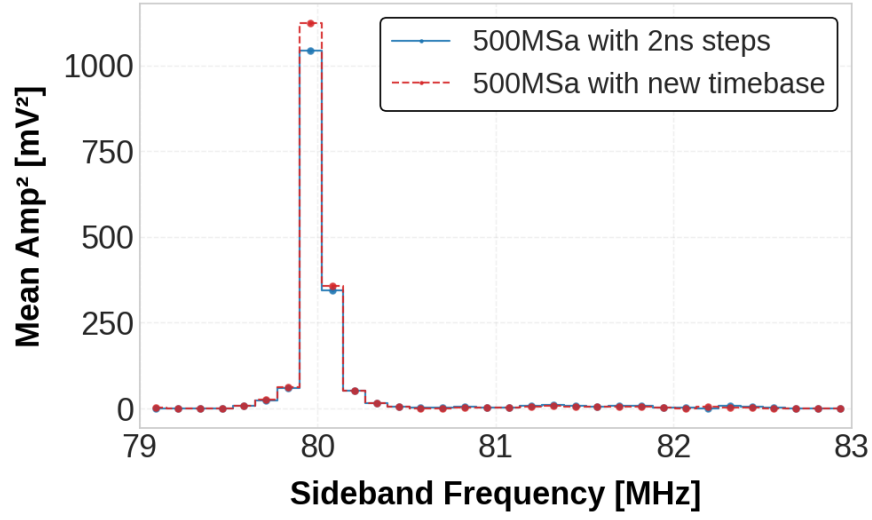
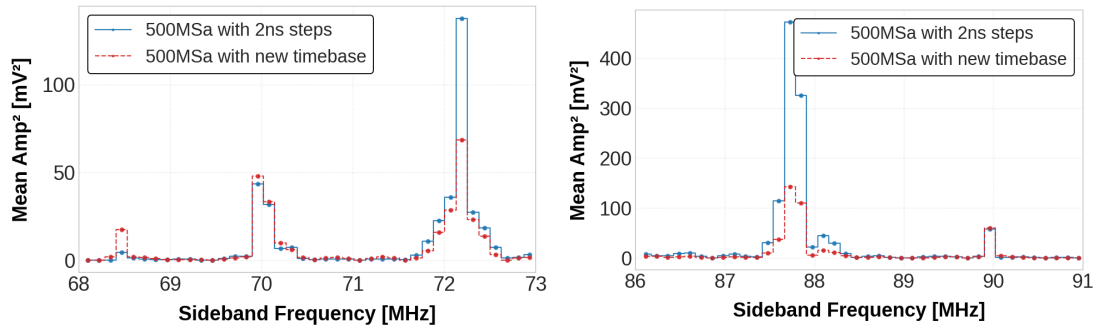


Figure 51: Power vs frequency at 80 MHz carrier with uniform and corrected time base.



(a) Power vs frequency at 70 MHz sideband with uniform and corrected time base. (b) Power vs frequency at 80 MHz sideband with uniform and corrected time base.

Figure 52: Power vs frequency for 70 MHz and 90 MHz sideband frequencies

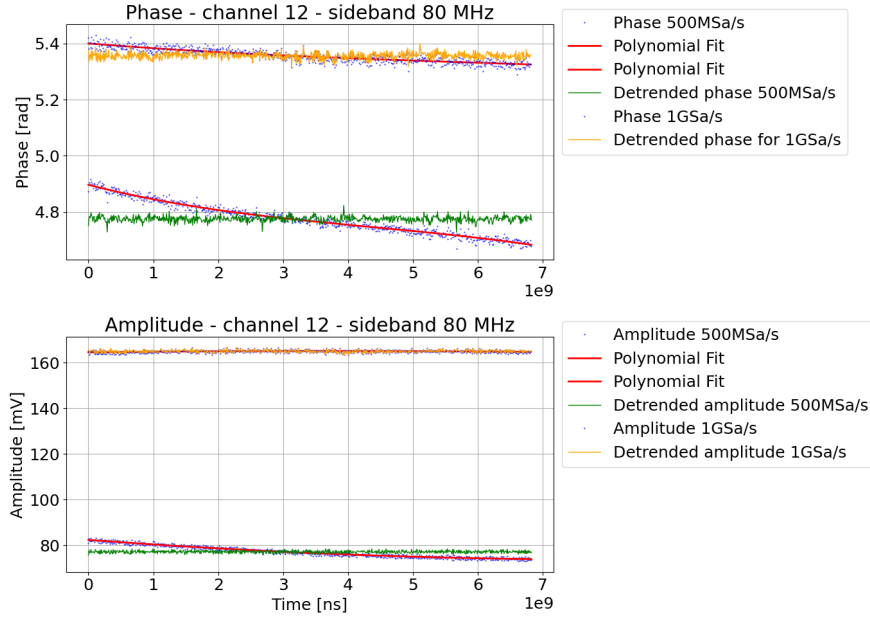


Figure 53: Phase and amplitude from IQ-demodulation described in subsection 4.3 for Pixel 14 and carrier frequency 80 MHz

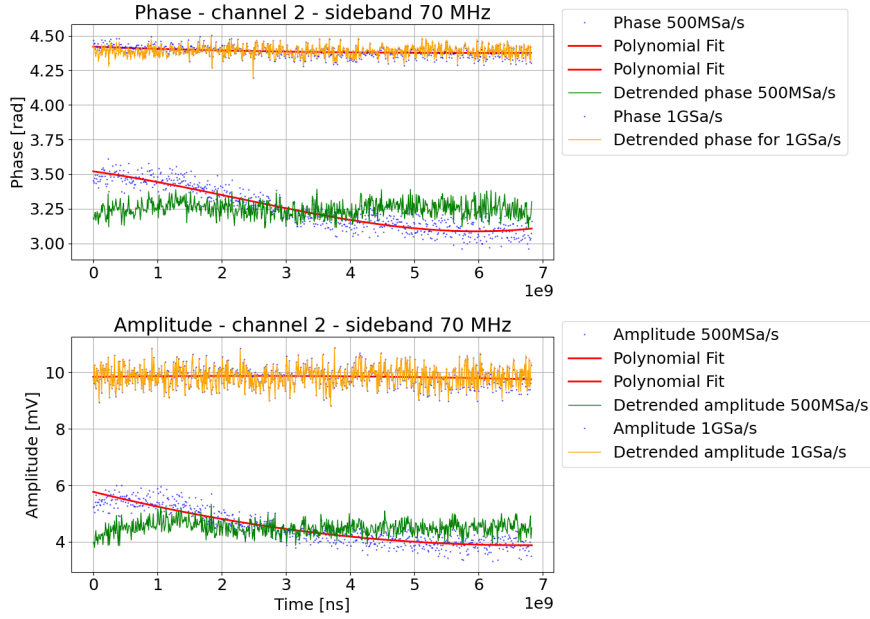


Figure 54: Phase and amplitude from IQ-demodulation described in subsection 4.3 for Sideband 70 MHz

For 500 MSa/s, the time array was build with 2 ns differences between consecutive times. These analyses were then combined to visualize the effect of the 500 MSa/s sampling rate.

The resulting phase and amplitude resolutions are plotted in orange for 500 MSa/s with the uniform time base, in green with the new time base and blue for 1 GSa/s. The

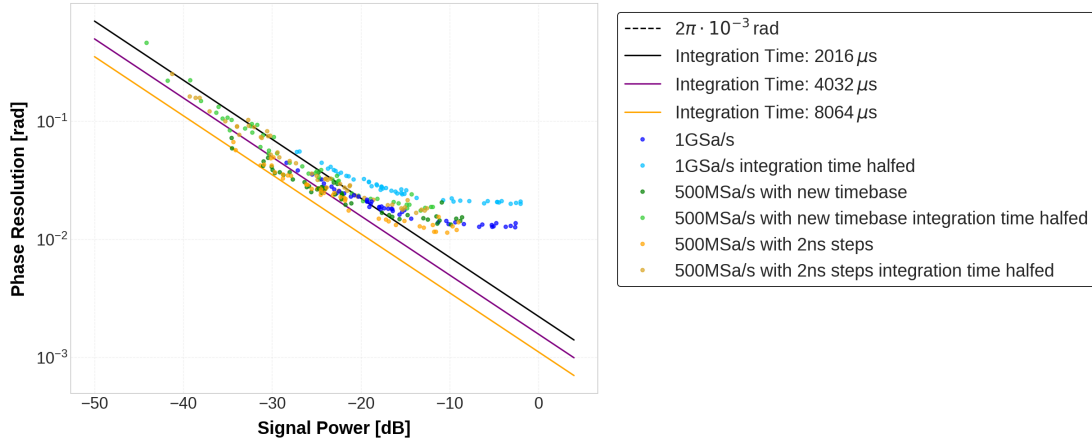


Figure 55: Phase resolution for data taken with the laser setup and TARGET ASICs at 500 MSa/s

lighter colors represent the same data with half the integration time. So for the 1 GSa/s this is 2016 μ s, for 500 MSa/s the same amount of samples results in an integration time of 4032 μ s. The phase resolution is plotted over the signal power, which is calculated as the mean of the amplitude of that sideband and pixel and then converted to dBm. This plot is shown in Figure 55. The noise estimation for 2016 ns, 4032 ns and for 8064 ns are plotted. It can be seen, that the phase resolution is decreased by halving the sampling rate Figure 55. For this the dark blue 1 GSa/s has to be compared to the light green 500 MSa/s curve, since for those the integration time is the same. When increasing the integration time to 8064 ns the amount of samples for 500 MSa/s is the same as with 1 GSa/s and an integration time of 4032 ns. This decreases the phase resolution. So we can have the same amount of samples in double the time with better phase resolution. The data for both sampling rates fits perfectly to the noise estimation up until approximately 1×10^{-2} rad. There the data describes a curve and flattens. This limit can be explained by the noise of the laser setup which consists of mirrors and lenses and posts. These components are vibrating at their own frequency and therefore influence the phase of the laser. The amplitude resolution for the same dataset is plotted in Figure 56. This was calculated as the standard deviation from the mean amplitude of each pixel. The amplitude resolution is similar for all analysis methods, showing no influence of the sample rate, time base or integration time. This plot proves that although we sample at a lower sampling rate, which decreases the phase resolution at the same integration time, we can overall improve the phase resolution through sampling rate reduction since we can extend the integration time. It also shows the time base correction having a very low effect on the phase resolution which means the uniform time base is enough to create minimal phase resolution for the implementation in the fiber-array based phase camera.

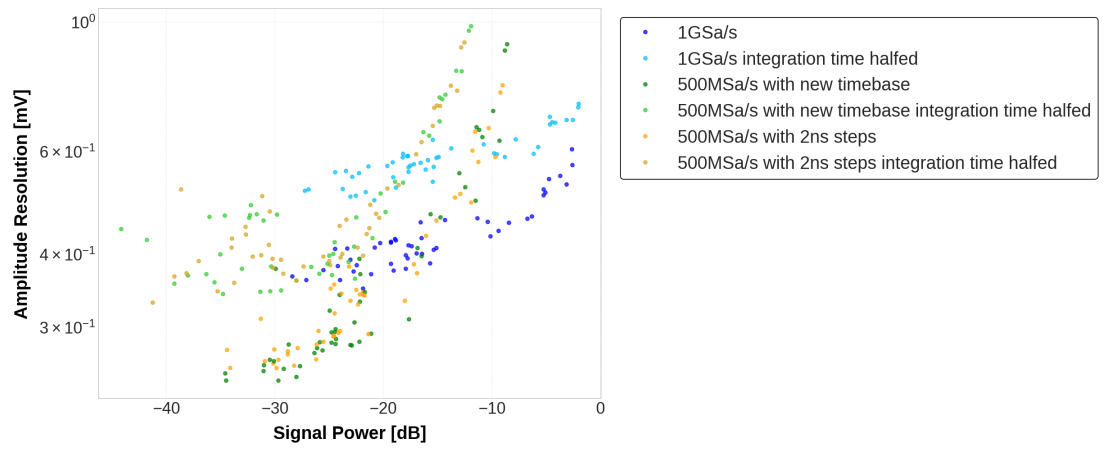


Figure 56: Amplitude resolution for data taken with the laser setup and TARGET ASICs at 500 MSa/s

11 Conclusion

The thesis goal was to characterize and optimize TARGET ASICs as a scalable solution for a fiber-array based phase camera for alignment control in the Einstein Telescope. The thesis analyzed different methods to improve the phase resolution for accurate phase reconstruction. Synchronization analyses showed a significant increase in temporal stability for wavefront reconstruction and alignment of multiple TARGET modules, while clock jitter was reduced by comparing different GPS clocks. The improvement from the dc-transfer function showed that this baseline correction is necessary for accurate phase reconstruction. An additional improvement was achieved by lowering the sampling rate, which increased the integration time, proving that the integration time is a key parameter in optimizing the phase resolution, despite the trade-off of lower sampling rate. Timing optimizations were achieved through timing parameter analysis and non-uniform time base generation. Additional studies for full buffer readout were started but no conclusive results were achieved yet and therefore remain the topic of future work. These findings support that the TARGET ASICs are suitable for the fiber-array based phase camera readout, and benefit from the calibration and timing optimizations. This shows concrete optimizations for the readout of the fiber-array for precise alignment control in future gravitational wave detectors.

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Declaration of Originality

I, Niklas Kotschi, student registration number: 22727540, hereby confirm that I completed the submitted work independently and without the unauthorized assistance of third parties and without the use of undisclosed and, in particular, unauthorized aids. This work has not been previously submitted in its current form or in a similar form to any other examination authorities and has not been accepted as part of an examination by any other examination authority.

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